The Topology of Slicing: Locating and Removing Forced Interruptions with a Map Chain

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Abstract

Some interruptions of a tool path are chosen by the planner; others are forced by the slicer's structure. We describe a slicer as a chain of maps, from a source through the part and the machine to the built part, and use the chain to locate forced interruptions and design around them. The obstruction is topological: a plan that must visit c path components is interrupted at least $c - 1$ times, and the chain identifies the link that supplies them. At the source, every reviewed solid-filling slicer traverses separate layers or surfaces, so its stops grow with resolution. Replacing the source by one connected volume, traversed by a three-dimensional Hilbert curve and carried into the part by optimal transport, gives a continuous path with the prescribed occupation measure on convex parts. In a computational prototype, it makes no stops on convex parts, where a layered raster's stops roughly double with resolution; on non-convex parts, a source split into near-convex pieces trades a fixed number of transitions for bounded excursions. At the machine, the components are the posture sheets of the robot's safe configurations, and a slicer that fixes the tool path first inherits the posture switches that path forces. Reversing the construction order gives kinematics-first slicing, in which the sheets guide the source split. On a three-link planar arm among obstacles, it makes the single posture reset required by the bound in a sampled availability model, while a geometry-first baseline makes two. The two constructions serve as demonstrations of the framework: one counting argument bounds both stops and posture switches, and in both a source split matched to the components attains the minimum.

1 Introduction

A slicer turns a part into machine motion, and that motion is rarely one continuous stroke. The flow stops between layers, and a robot may have to change posture partway through. Some of these interruptions are chosen by a planner and could be planned away; others are forced by the structure of the slicer, and no planner can remove them. This paper is about the forced ones: where they come from, which stage of the slicer forces them, and how to design a slicer that avoids them. Figure 1 shows a tool path built this way.

On a simple cylinder, a layered raster must stop between layers, and the count roughly doubles each time the resolution doubles (Section 4). Each stop of the material flow can leave a seam, and in some processes a stop is especially costly: it can cut reinforcing fibre in continuous-fibre printing [1], cause a restart defect in wire-arc deposition [2], or weaken the bond in concrete printing [3]. Better paths within each layer remove some stops, but not those between layers. A robot shows the same pattern one stage later. When obstacles split its safe postures into separate families, some tool paths cannot be followed without a posture reset, however good the inverse kinematics.

Thesis: forced interruptions are topological. Both cases have one cause. A continuous stretch of motion stays within one path component of the space it moves in. When the plan must visit c components, it is interrupted at least $c - 1$ times, whatever order it uses. Stops

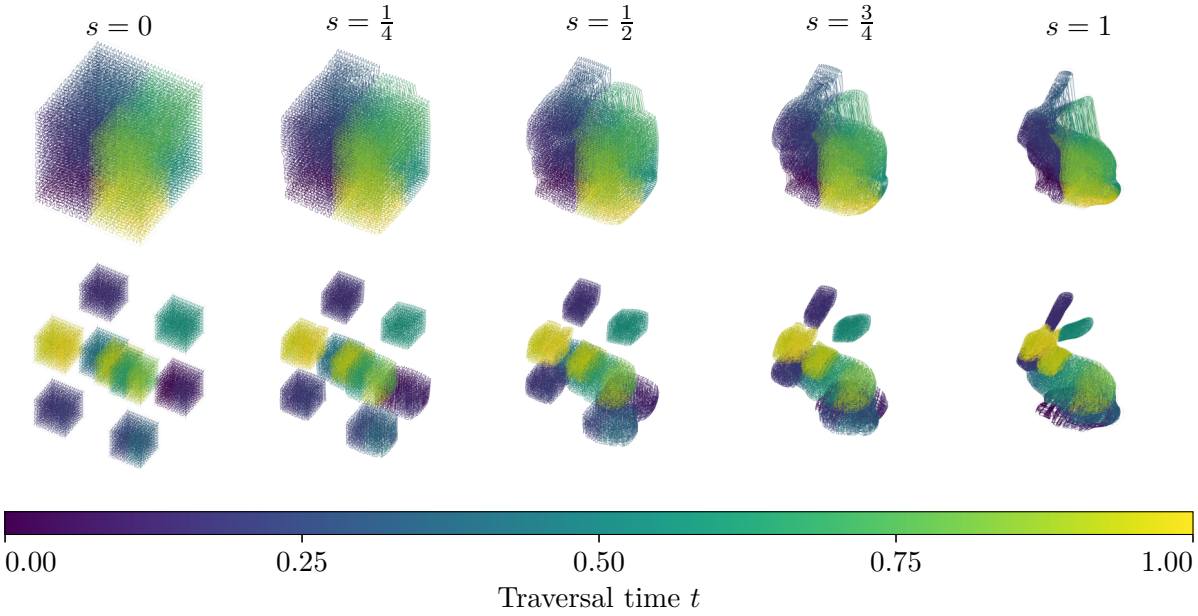


Figure 1: The first application of the framework: a connected volume, not a stack of layers, as the source. Top: a Hilbert path through a cube (32 768 cells) is moved into the Stanford bunny by optimal transport; intermediate shapes are drawn for illustration. Bottom: a decomposed source of nine cubes, each transported onto one piece of the bunny. Colour indicates traversal time. The visiting order comes from the cubes, and the shape, including the ears, from the map alone.

come from a source made of separate pieces, the layers. Posture resets come from a fibre space of safe configurations made of separate sheets. The bound counts path components, which is why we call the subject the topology of slicing. Locating an interruption therefore means finding the link that supplies the components. Removing it means changing that link, or the order in which the links are built, and then meeting whatever guarantees the change leaves as obligations.

The map-chain framework. We propose to treat a slicer as a chain of maps, from a source representation through the part and the machine to the built part (Section 2). Each link carries its own guarantees, which compose. Each forced interruption can therefore be traced to the link that supplies its components, and no change to the other links can remove it. This is also a design method. Reading the reviewed slicers link by link shows which choices they share. Replacing one of those choices, or changing the order in which the links are built, defines a new construction; the chain states which guarantees survive and which become obligations. Within the slicing literature we review, two shared choices force interruptions, at two different links. We construct an algorithm from the alternative to each; the two algorithms are demonstrations of what the framework makes visible, not its full extent.

First application: changing the source. Every reviewed solid-filling slicer traverses a source of separate layers or surfaces, including methods that represent the solid as a volume, such as trivariate parameterisations, scalar fields, frame fields, and implicit neural fields [4–7]: each still cuts the volume into surfaces and visits them in turn. Separating the source from the traversal makes the alternative visible: a source that is one connected volume, traversed directly. The *transported Hilbert path* constructs it (Section 3). A three-dimensional Hilbert curve orders a cube, and optimal transport carries the cube into the part. On convex parts the path is continuous and has exactly the prescribed occupation measure, and a semi-discrete computation gives every waypoint an equal share of the target mass, up to solver tolerance.

Containment on non-convex parts remains an obligation of the geometry map; decomposing the source into near-convex pieces meets it up to each piece's hull gap. Hilbert curves and optimal transport are not new; the construction is their use as source, traversal, and geometry map. Here the interruption lies in the source, and so does its removal.

Second application: changing the construction order. Every reviewed slicer builds the tool path first; robot-aware methods then plan machine motion with that path, or its geometric elements, as input. The path then fixes the posture switches, because it fixes how often the tool passes between regions that only one sheet reaches. Separating the geometry map from the lift makes the alternative visible: build admissible machine motion first and derive the tool path from it through forward kinematics. *Kinematics-first slicing* constructs it (Section 5). Static feasibility then holds by construction, while coverage and the prescribed deposition density become obligations. The robot's posture sheets guide how the source is split, and patches with continuous safe lifts, each visited once, attain the minimum number of posture switches. The bound is proved by the same counting argument as the bound on stops, now applied at the machine link. Configuration-first coverage is studied in robotics [8]; the framework connects it to the deposition measure and to source decomposition.

Scope of this investigation. The two applications are preliminary demonstrations, and each stops at a clearly drawn boundary. Their experimental evidence is computational and differs between them. The volumetric construction is evaluated computationally on five parts and two decompositions (Section 4). It is realised at finite Hilbert order, on polygonal parts, and it stops at the geometric tool path: no lift, machine execution, or deposition is modelled, and no part has been printed. The process costs of stops cited above motivate the example; we do not demonstrate them. The decomposition results are sampled measurements from one experiment, not a proof of containment; the crossings into neighbouring pieces leave local density and overlap unmeasured, and the k piece paths are not yet sequenced. The kinematics-first construction is evaluated on one three-link arm, obstacle layout, and target at Hilbert orders two to four (Section 5.2), so one reset against two is not a general rate, and the timings are specific to one machine and implementation. Admissibility there is static: joint rates, dynamics, and the process Θ remain unchecked. Sheet separation and the optimality of the switch counts hold only in a sampled availability model; sampling rules out neither a connection through an unsampled passage nor availability at unsampled radii. Posture resets are counted, but the physical transition between postures is neither planned nor certified, and no safe physical transition between the two families is claimed. Each of these boundaries is also a point where the chain says what the next step must supply; Section 6 returns to them.

Appendix A formalises the framework and proves its composition and transition results; Appendix B proves the guarantees of the volumetric application; Appendix C gives the volumetric experimental methods and supporting results; Appendix D gives the kinematics-first proof, experimental methods, and supporting results.

Research prototype and reproducibility. The research prototype, saved reference results, and figure-generation code are available at `https://github.com/pobol2/Topology-of-Slicing`.

2 The map chain as a slicer design framework

A conventional slicer bundles several decisions into one algorithm. Separating them lets us compare slicers one decision at a time, trace each limit to the decision that causes it, and construct new slicers by making a decision differently.

2.1 The map chain

Figure 2 reads a slicer as a sequence of decisions. A slicer first chooses a *source* $\mathcal{X}$, the space in which the path is designed: a stack of slices, a family of surface patches, or a volume. A *traversal* ξ then decides the order in which the source is visited over normalised deposition time $t \in [0, 1]$. A *geometry map* Φ places each source point in the part $\Omega \subset \mathbb{R}^3$, which produces the tool path. Finally, a *lift* Ψ_g turns the tool path into motion in the machine's configuration space $\mathcal{Q}$, and the *process* Θ turns that motion into the built part $\widehat{\Omega}$:

$$x(t) = \Phi(\xi(t)), \qquad q = \Psi_g[x], \qquad \widehat{\Omega} = \Theta[q]. \tag{1}$$

The traversal and the geometry map act point by point. The lift and the process act on whole paths, hence the brackets, because a machine's configuration and the material it has laid down depend on what came before. The subscript g records the usual order: geometry first, machine second.

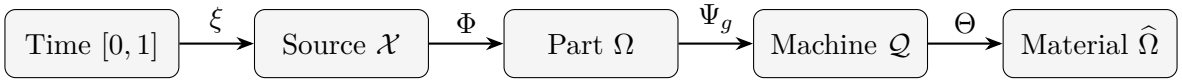


Figure 2: The map chain. Traversal and geometry act pointwise; the arrows into machine motion and material act on whole paths.

The chain also decides how much material goes where. Suppose the traversal spends its time in the source according to a distribution $\mu_{\mathcal{X}}$, and the geometry map sends $\mu_{\mathcal{X}}$ to the prescribed distribution μ_Ω on the part. Then the tool path spends its deposition time according to μ_Ω. Writing $f_{\#}\mu$ for the image of a distribution μ under a map f, and λ^1 for uniform time,

$$\xi_{\#}\lambda^1 = \mu_{\mathcal{X}}, \quad \Phi_{\#}\mu_{\mathcal{X}} = \mu_\Omega \quad \Longrightarrow \quad x_{\#}\lambda^1 = \mu_\Omega. \tag{2}$$

With constant flow, time shares are material shares. The identity therefore states what it means for a path to cover the part with the right amount of material. Graded density needs no separate link: any chain can carry it through $\mu_{\mathcal{X}}$ or μ_Ω.

A planar slicer illustrates every link. Its source is the set of separate cross-sections $\mathcal{X} = \bigsqcup_{k=1}^{N}(D_k \times \{k\})$. Its traversal zigzags within each slice and visits the slices in order. A gantry with tool offset o then uses

$$\Phi(u, v, k) = (u, v, z_k), \qquad \Psi_g[x](t) = x(t) - o, \tag{3}$$

and the process turns the resulting motion into deposited material. The resulting distribution lives on N slices, so the process must fill the gaps between them. Appendix A.1 gives the precise spaces, the deposition-time convention, and the formal definition (Definition A.1); there, a finite raster's source measure is the occupation measure of the zigzag rather than the ideal area measure on each slice.

The chain is useful because guarantees attach to links. Exact distributions, continuity, containment, and error bounds compose from one link to the next; Appendix A.2 states the composition inequalities with proofs (Proposition A.2) and tabulates the resulting guarantees (Table 5). Each failure therefore has an address: a disconnected source forces transitions, a geometry map can tear the path or leave a non-convex part, a lift can switch branches near a singularity [9], and the process can leave gaps. This paper stops before the process: it does not model how material is deposited. With this decomposition, each existing method becomes a combination of independently chosen links.

Designing with the chain. The same decomposition gives a procedure for deriving new slicers. Read an existing construction link by link and identify a choice that it holds fixed. Then either replace that choice, or change the order in which the links are constructed. Finally, use the composition rules to determine which guarantees survive the change and which become obligations for the remaining links. Sections 3 and 5 apply this procedure to the two choices that the review below exposes.

2.2 Existing slicers as link choices

Table 1 places the reviewed slicer families in the chain and can be read in two directions. Along each row, it records which links a family changes relative to the planar chain; this is where the families differ. Down the source column, it records the source as it is actually traversed; this is where families that look very different turn out to agree. Adaptive layer height changes the spacing of the slices. Curved-layer slicers bend them through the geometry map. Multi-axis slicers change the lift, and within-layer continuous paths change the traversal inside each slice. Families often combine changes: S^3-Slicer, the scalar-field method, and the neural slicer target multi-axis machines and so also change Ψ_g [5, 10, 11].

Family	Source $\mathcal{X}$	Links changed	Order	Examples
Planar (reference)	N flat layers	–	Geom.	[12]
Adaptive layer height	N flat layers, variable spacing	$\mathcal{X}$	Geom.	[13–15]
Within-layer continuous	N flat layers	ξ	Geom.	[16–18]
Curved layers	N curved layers	Φ	Geom.	[10, 19]
Multi-axis	N layers or surfaces	Ψ_g	Geom.	[9, 20, 21]
Scalar-field	N level surfaces	$\mathcal{X}$, Φ	Geom.	[5]
Trivariate	Parameter volume, visited surface by surface	$\mathcal{X}$, Φ	Geom.	[4]
Neural mappings	N learned curved layers	Φ	Geom.	[7, 11]
Surface paths	N surface patches	$\mathcal{X}$, ξ	Geom.	[22]
Vase mode	One surface (a single wall)	$\mathcal{X}$, ξ, Φ	Geom.	[12]
Transported Hilbert path	One solid volume, or k volumes if decomposed	$\mathcal{X}$, ξ, Φ	Geom.	This paper, Sections 3–4
Kinematics-first	c patches, one per posture sheet	$\mathcal{X}$, Ψ_k	Kin.	This paper, Section 5

Table 1: Slicer families as choices of links, relative to the planar chain. The source column is the source as it is actually traversed; the order column records whether the tool path (Geom.) or the machine motion (Kin.) is constructed first. Every reviewed family above the rule builds geometry first, and every solid-filling one traverses separate layers or surfaces. The table is representative, not exhaustive. Below the rule are the two constructions of this paper: the first changes the source; the second changes the construction order, and its map Ψ_k into configurations takes the place of both Φ and Ψ_g.

Two choices remain shared across the reviewed families. All of them build geometry first. And every solid-filling family among them traverses its source as N separate pieces. A volumetric representation does not change this second choice. The trivariate method parameterises a volume but still visits it surface by surface [4], and volumetric frame fields [6] and implicit neural fields [7] likewise describe the solid as a volume but print it as layers or surfaces. Conversely, the many methods that remove travel by a better order, such as Fermat spirals and the continuous planners of wire-arc manufacturing [16, 23, 24], improve the traversal inside each piece of the source, not the source itself. These shared choices leave two alternatives absent from the reviewed slicing methods: traversing a connected volume, and constructing the machine motion before the tool path. The first concerns the source, whose disconnection carries a cost that no other link can pay

back, as the next subsection shows; the second concerns the order of construction (Section 5).

2.3 The hidden limit: pieces force stops

Let the source have N separate pieces, each receiving positive deposition time. Any traversal that is continuous except at finitely many times then makes at least $N-1$ *transitions* between pieces, in whatever order it visits them; Appendix A.3 proves this bound (Proposition A.5) and its consequence for separated images. This is the counting argument of Section 1: a better order shortens the moves but cannot reduce their number. Section 5 applies it again at the machine link.

A transition is an event in the source; its effect on the machine depends on the later links. When the mapped pieces do not touch, a transition becomes a break of the tool path. A break forces a *stop*: a break occupies no deposition time, so the machine crosses it with a travel move during which the flow is switched off. For layers, every transition is a stop, because distinct layers lie at least the layer spacing h apart (Corollary A.6).

The stop count of a layered slicer therefore grows with resolution. A part of height H needs at least $H/h - 1$ stops: a 100 mm part has 500 layers and at least 499 stops at a layer spacing of 0.2 mm, and at least 999 stops at 0.1 mm. Disconnected cross-sections add more. No reviewed family in Table 1 lowers this count. Continuous within-layer paths act inside each piece, curved layers and surface patches are still separate pieces, and multi-axis lifts act after the source is fixed. Vase mode escapes with a connected surface, but it prints a single wall [12].

This diagnosis is the first step of the design procedure: since stops originate in the source, the source itself must change. The simplest replacement for a solid is a single connected volume. The composition rules then say what the change costs: a connected source removes forced transitions only if a traversal can visit it continuously and a geometry map can carry it into the part without tearing it apart again. Section 3 constructs the first algorithm by meeting these two obligations.

3 First application: changing the source

The first application follows the design procedure of Section 2 step by step. A disconnected source forces transitions (Section 2.3); choosing the unit cube, a connected volume, removes that structural cause. The composition rules then state what the remaining links must supply. The tool path $x = \Phi \circ \xi$ is continuous only if both the traversal and the geometry map are, and by (2) it has the prescribed occupation measure only if

$$\xi_{\#}\lambda^1 = \lambda^3, \qquad \Phi_{\#}\lambda^3 = \mu_\Omega, \tag{4}$$

where λ^3 is uniform volume on the cube. The traversal must therefore be a continuous, measure-preserving curve through the cube, and the geometry map must be a continuous map that pushes uniform volume onto μ_Ω. A space-filling curve meets the first obligation exactly, and optimal transport meets the second, with continuity under the convexity assumptions below. The resulting *transported Hilbert path*, or transported path for short, gives each map a single job: the traversal orders, and the geometry map shapes. The construction ends at the tool path: the lift and the process remain abstract links of the chain.

3.1 Construction: Hilbert orders, transport shapes

The traversal is the three-dimensional Hilbert curve $\mathcal{H}_3$ [25–27]. It visits every point of the cube continuously, spends equal time in equal volumes, and keeps points visited at nearby times close together. At a finite order m it visits the centres $u_1, \dots, u_n$ of $n = 8^m$ cells, consecutive cells sharing a face [28]; Appendix B.1 establishes its measure preservation, locality, and finite-order estimates. Here the geometry map Φ is chosen as the optimal-transport map. Among all maps

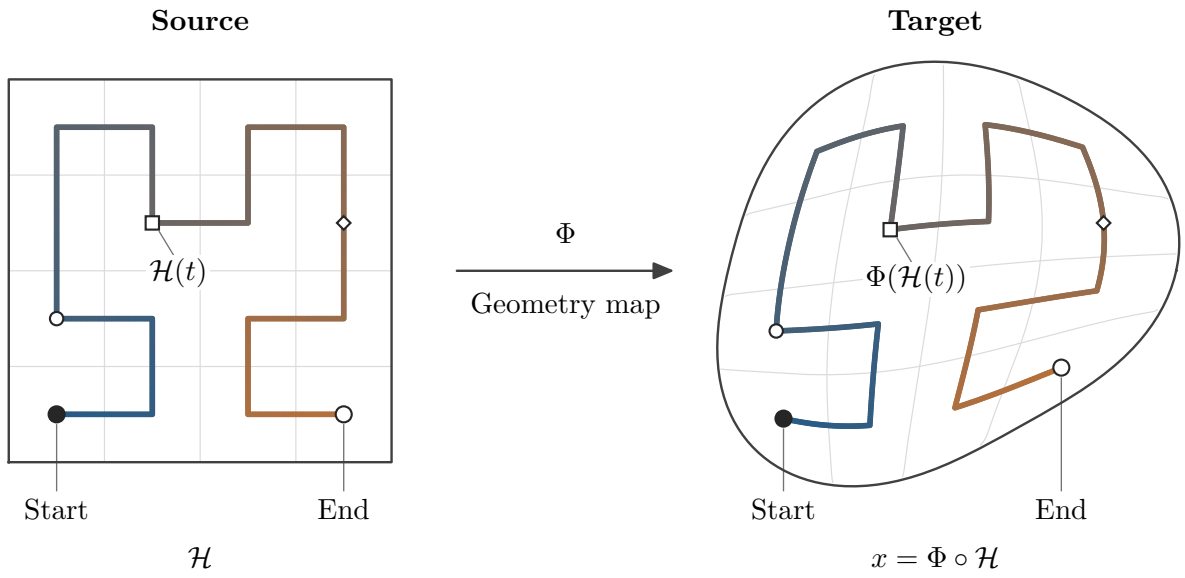


Figure 3: Hilbert orders, transport shapes. A numerical two-dimensional illustration of the volumetric construction: an order-two Hilbert traversal $\mathcal{H}$ in the source square (left) is carried into the target by a smooth approximation of the geometry map Φ (right), computed from an entropically regularised quadratic optimal-transport solve. Matching colours and marker shapes indicate corresponding traversal times; the visiting order is inherited from the source.

that send uniform volume in the cube to the target distribution μ_Ω, it minimises the mean squared displacement [29, 30]. The transported path is

$$x(t) = \Phi(\mathcal{H}_3(t)). \tag{5}$$

Composition gives $x_\# \lambda^1 = \Phi_\# \lambda^3 = \mu_\Omega$; when transport is continuous, the composed path is continuous as well. Figure 3 illustrates the construction numerically in two dimensions, and Figure 1 (top) illustrates the construction on the Stanford bunny. Its distinction is to transport a traversed volume while preserving the prescribed occupation measure: earlier manufacturing uses of Hilbert curves traverse layers or deposition surfaces [31, 32], while curved-layer, multi-axis, and neural slicers deform layers [10, 11, 19].

3.2 Guarantees on convex parts

The Hilbert curve is continuous, so the transported path is continuous wherever the transport map is. Optimal transport always delivers the prescribed distribution, but it need not be continuous: a map that sends a cube onto a shape with a hole may have to tear. Convex targets avoid this. If the part is compact and convex with nonempty interior, and the target density is bounded above and below by positive constants, the transport map is Hölder continuous on the closed cube [33, 34]; Appendix B.2 explains why these assumptions suffice. We call these assumptions the *convex setting*.

In the convex setting, the ideal transported path is continuous and has the prescribed occupation measure, $x_\# \lambda^1 = \mu_\Omega$; Appendix B.3 derives this (Theorem B.1(ii)) together with the finite-order bounds below. Physically, the nozzle never has to break off and travel: the path has no stop, and the time it spends in each region matches the prescription.

At a finite Hilbert order m, the guarantee becomes a bound. Waypoints $\hat{x}_i$ that lie within ε of $\Phi(u_i)$ satisfy, for constants $\beta \in (0, 1]$ and C_Φ (Theorem B.1(i)),

$$\text{covering radius} \le C_\Phi 2^{-\beta m} + \varepsilon, \qquad \text{step length} \le C_\Phi 2^{-\beta m} + 2\varepsilon. \tag{6}$$

The two terms belong to different links. A finer Hilbert order shrinks the first; only a more accurate geometry map shrinks the second. The constants are not known for a computed map, which is why the computation below supplies a bound of its own.

Outside the convex setting, the guarantee is absent rather than refuted: transport still delivers the prescribed distribution, but no theorem makes it continuous. Implementing the construction also requires computing the map, which brings its own certificates and its own gaps.

3.3 Implementation: semi-discrete transport

The waypoints are computed by semi-discrete optimal transport between the part and the finite Hilbert curve [35–37]. The order-m curve is the discrete measure $\nu_m = n^{-1} \sum_i \delta_{u_i}$, one unit of time per site, and the part carries the continuous target $\mu_\Omega = \rho\, dy$, which may be graded. For weights $\psi \in \mathbb{R}^n$, the *Laguerre cell* of site u_i is the part of Ω that is power-nearest to it,

$$L_i(\psi) = \{y \in \Omega : \ \|y - u_i\|^2 - \psi_i \le \|y - u_j\|^2 - \psi_j \text{ for all } j\}. \tag{7}$$

The cells always tile Ω. The solver adjusts ψ until every cell carries its site's share of the target, up to solver tolerance,

$$\mu_\Omega(L_i(\psi)) = \tfrac{1}{n} \qquad (i = 1, \dots, n), \tag{8}$$

which makes the map sending each L_i to u_i the optimal transport from μ_Ω to ν_m [30, 35]. Each Hilbert site is finally replaced by the density-weighted centroid of its cell,

$$\hat{x}_i = n \int_{L_i(\psi)} y\, \rho(y)\, dy, \tag{9}$$

and the waypoints are visited in Hilbert order. A Hilbert site thus labels a cell of the part rather than serving as a waypoint, and the waypoint is the centre of mass of that cell. On the cube with uniform density, each cell is the Hilbert cell and $\hat{x}_i = u_i$: the construction is the identity, as it should be.

This construction controls, cell by cell, what the ideal map is meant to control; Appendix B.4 proves its distribution, order, and containment properties (Proposition B.2) and the hull-gap bounds used below.

- *Cell mass.* Every waypoint stands for the same mass $1/n$ of the target, matching the equal time that the Hilbert curve spends in each cell. No target sample is drawn and none is deleted.
- *A computable distribution bound.* Sending each cell's mass to its own centroid moves no mass farther than the largest cell diameter $D_m = \max_i \operatorname{diam} L_i(\psi)$, so the waypoints match the target to within $W_\infty \le D_m$. Unlike ε in (6), D_m is measured for every computed path.
- *Order.* Each centroid stays power-nearest to its own site, so the waypoints inherit the Hilbert labels without reordering.

Three things are not controlled, and the third becomes the turning point of the construction. First, how fast $\hat{x}_i$ approaches $\Phi(u_i)$ is not certified, so the step bound in (6) remains conditional. Second, equal cell mass is not waypoint-density accuracy: regions thinner than a cell are not resolved (Proposition B.2(iii)). Third, equal mass says nothing about containment. On a convex part each clipped cell is convex, so every centroid and every segment between consecutive centroids lies in the part. On a non-convex part, two mechanisms break this: a cell straddling a reentrant corner is clipped into disconnected portions whose combined centroid can lie in the empty region between them, and two cells with centroids inside the part can be joined by a segment that crosses a hole. Figure 5 (Section 4) shows both on computed paths.

3.4 Repair: a volumetric source in several pieces

The chain assigns this failure to the geometry map, not the source: the cube is still connected, and the excursions arise where cells are clipped by holes and reentrant edges. The repair therefore

stays within the first application: it restores the geometry map's convexity assumption, or nearly so, while keeping the source volumetric. Instead of one cube, the source becomes k cubes, each transported onto a convex or near-convex piece of the part. Every piece is still filled by a path that runs through its volume; what changes is that the part now receives several piece paths instead of one. How far each piece may be from convex is measured by a single quantity.

A geometric guarantee for near-convex domains. For a compact domain K, let $\delta_K = \sup_{z \in \mathrm{conv}(K)} \mathrm{dist}(z, K)$ be its *convex-hull gap*: how far its convex hull reaches beyond it. Centroids of cells contained in K, and straight segments joining them, lie in its convex hull, hence within distance δ_K of K (Proposition B.3). This bound needs no transport regularity and recovers exact containment when K is convex. It bounds excursion depth only.

The decomposed source. Given a partition of the part into k convex or near-convex pieces $\Omega_1, \ldots, \Omega_k$, the source is k cubes, one per piece, and the Hilbert traversal and semi-discrete transport are applied independently to each (Figure 1, bottom). Each piece receives its share $w_j = \mu_\Omega(\Omega_j)$ of deposition time, and its path leaves it by at most its own hull gap δ_j, with exact containment for convex pieces (Corollary B.4).

Two kinds of excursion must now be kept apart. A piece path can leave its *assigned piece*, by at most δ_j; such an excursion may enter a neighbouring piece and so remain inside the part. Or it can leave the *whole object*, which is the failure that decomposition is meant to remove. The hull-gap bound controls the first directly; it controls the second only where the region within δ_j of a piece lies inside the part.

The remaining tradeoff. Decomposition reintroduces separate pieces into the source, so Proposition A.5 forces at least $k - 1$ transitions between piece paths. Unlike the N layers of a layered slicer, k is set by the partition of the part, not by the Hilbert order, so the count does not grow as the resolution is refined. Sequencing the pieces, and hence executing these transitions, is outside the scope of this paper.

4 Evaluating the first application

The framework makes three predictions about the first application, each naming a link: changing the source removes the transitions between layers; the failures that remain belong to the geometry map; and decomposition repairs them up to each piece's hull gap. A computational demonstration tests them in that order. Its scope is stated in Section 1.

The implementation computes the semi-discrete map of Section 3.3 with the solver of Geogram [37, 38], on parts meshed into tetrahedra [39], solving coarse to fine. The five parts are a cylinder, a sphere, a solid torus, an L-bracket with a reentrant edge, and a complex part consisting of a block with several holes and a boss (genus three). Every run, on these parts at orders $m = 4, 5, 6$ (4096 to 262 144 waypoints), met the equal-mass condition to a relative tolerance of 10^{-6}. A stop is counted wherever a path would have to be executed as travel: for the layered raster baseline, each travel move; for the transported path, each maximal *exterior run* of segments that leave the part. Steps longer than five times the mean are reported separately as *jumps*. Appendix C.1 gives the meshing, solver settings, and acceptance checks; Appendix C.2 defines the metrics, evaluation samples, and the layered baseline; and Table 6 in Appendix C.3 reports all fifteen runs.

A connected source removes the stops on convex parts. This is the central result of the first application. Figure 4 shows the mechanism on a cube at order two; the counts below are for the cylinder. The decisive count is the number of stops between layers, which Proposition A.5

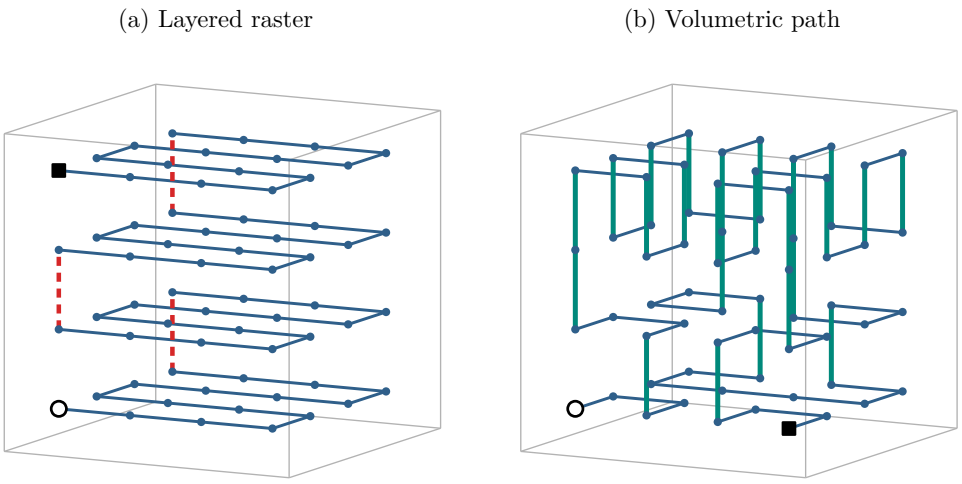


Figure 4: Layer changes on a cube at order two (64 waypoints). (a) Layered raster on the same cell centres: dashed red segments mark the three layer changes, each requiring a stop. (b) Computed transported path: it changes height throughout (teal segments mark mostly vertical steps), so there is no layer boundary to stop at, and it has no exterior run. The circle marks the start and the square the end. Stop counts under refinement are in Table 2.

forces. A layered raster makes 13, 27, and 57 of them at orders 4, 5, and 6, roughly doubling with the number of layers (Table 2). The transported path has no layers to stop between, and no exterior run at any tested order. That zero follows by construction on a convex part; the additional findings are that the path also has no jump, and that its longest step shrinks with the order (from 0.069 to 0.023 on the cylinder). Changing the source thus removes a limit that grows with every refinement of the layered chain.

	Layered raster		Transported path
Part	Between layers	All travel	Exterior runs
Cylinder	13 / 27 / 57	97 / 419 / 2029	0 / 0 / 0
Sphere	13 / 27 / 57	101 / 371 / 1533	0 / 0 / 0
Torus	5 / 11 / 23	69 / 271 / 1163	1 / 53 / 241
L-bracket	3 / 9 / 19	3 / 19 / 19	53 / 224 / 836
Complex part	5 / 11 / 23	37 / 155 / 586	95 / 323 / 1103

Table 2: Stops at Hilbert orders 4 / 5 / 6. Most raster travel lies within layers, where a better in-layer planner could remove some of it; only the stops between layers are forced.

The distribution is controlled cell by cell. Every accepted path assigns each waypoint the same target mass, so each waypoint stands for one Hilbert time block. On the convex parts the computable bound D_m roughly halves with each order, from 0.120 to 0.065 and 0.035 on the cylinder, and the covering radius follows, from 0.047 to 0.023 and 0.012. This is consistent with the $2^{-\beta m}$ term of (6), though three orders do not determine β.

On non-convex parts, the failures lie in the geometry map. As the framework predicts, the source stays connected and the failures come from non-convex clipped cells. On the torus at order four, every waypoint is inside the part, but one segment leaves the part across the hole. The L-bracket and complex part retain more exterior segments: at order four, 4.9% and 3.8% of their segments leave the part, and 97.3% and 98.8% of their waypoints lie inside (Table 6). Figure 5 shows both mechanisms at order four. On the L-bracket, cells near the reentrant edge

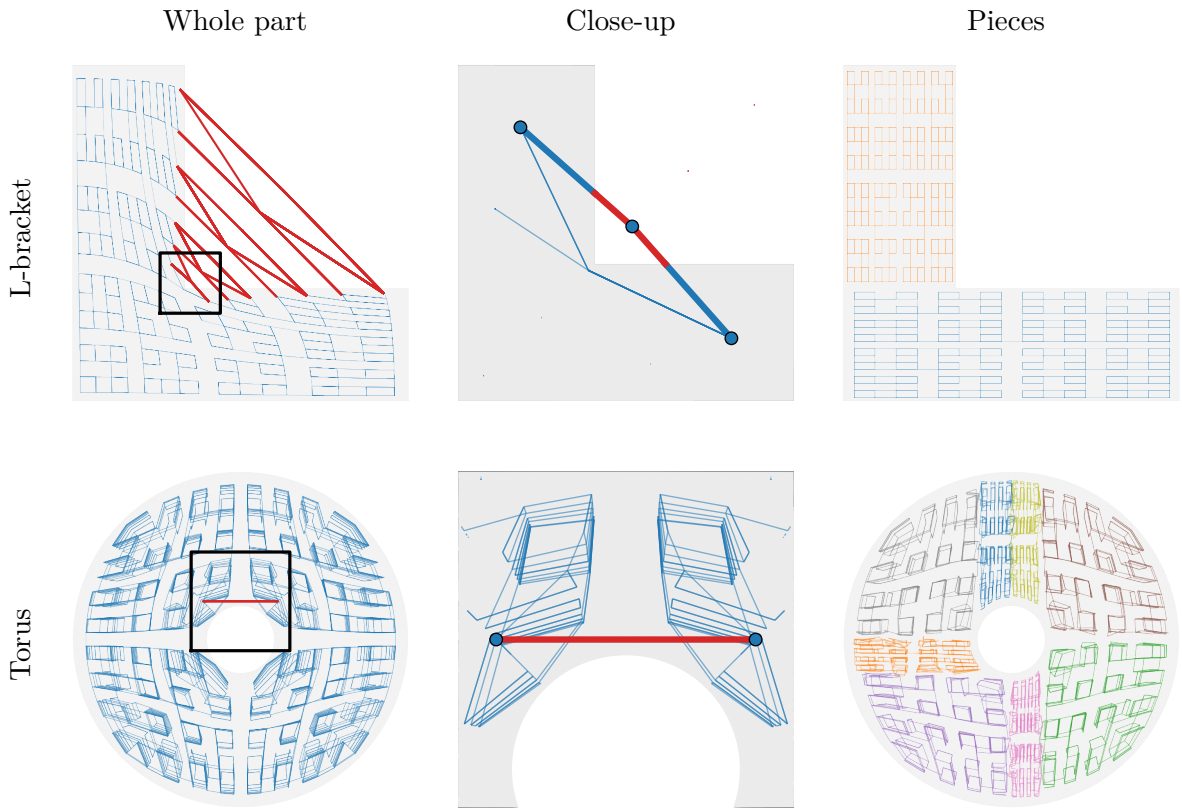


Figure 5: Containment failures and their repair, computed at order four (plan views; red marks the path outside the part). Left: whole-part paths (4096 waypoints). Middle: close-up of the boxed region; thick segments join consecutive waypoints (dots). On the L-bracket, an exterior run passes through a waypoint in the empty region beside the reentrant corner. On the torus, the single exterior segment joins two interior waypoints across the hole. Right: piece paths, with one colour per piece: the two convex boxes of the L-bracket at order four, and eight automatic torus pieces at order three (4096 waypoints in total, the whole-part budget; Appendix C.5). No piece-path segment leaves the part.

straddle the empty region between the arms and their centroids lie in it; on the torus, the one exterior segment joins two interior centroids across the hole, which is also how most exterior segments of the complex part arise. Appendix C.4 separates these two mechanisms by segment endpoints. The cost is real: the torus has fewer exterior runs than the raster has travel moves, but the L-bracket and complex part have more (Table 2).

Refinement reduces the excursions but does not remove them. Between orders four and six, the fraction of exterior segments falls from 0.049 to 0.012 on the L-bracket and from 0.038 to 0.007 on the complex part, but the deepest excursions persist (Table 6; Appendix C.4). Most cells shrink with the order, but those clipped by holes and reentrant edges stay large: on the L-bracket, D_m stays near 0.85. Refinement does not make the target convex. This persistence is what motivates decomposition.

Decomposition restores containment up to each piece's hull gap. We apply the same traversal and solver independently to the two boxes of the L-bracket and to an automatic partition of the torus into eight pieces. All ten local solves pass. The result has two parts, matching the two kinds of excursion of Section 3.4.

Against each *assigned piece*, the bound holds. The two convex boxes give full containment, as

Proposition B.3 guarantees (Figure 5, top right). The automatic pieces are not exactly convex: 5.21% of the torus piece-path segments leave their own piece, but every excursion stays below that piece's computed hull-gap bound (Table 7). Appendix C.5 gives the partition procedure, the numerical hull-gap evaluation, and the mass-weighted same-budget comparison.

Against the *whole object*, the excursions disappear in the torus comparison. The crossings above enter neighbouring pieces, not empty space. At the same budget of 32 768 waypoints, the whole-part path has 99 segments leaving the torus, while the piece paths have none, with similar coverage (Table 3).

Path / domain	Inside	Ext. seg.	Ext. length	Depth	Coverage	CV	Radius
Whole / torus	99.98%	0.30%	0.51%	0.0317	0.981	0.392	0.0244
Pieces / torus	100.00%	0.00%	0.00%	0.0000	0.978	0.405	0.0200
Pieces / assigned	98.40%	5.21%	3.11%	0.0232	–	–	–

Table 3: Torus at the same budget: one whole-part path and eight piece paths, 32 768 waypoints each. *Torus* measures against the whole object, *assigned* against each path's own piece. "Inside" is the percentage of waypoints inside the evaluation domain; "Ext. seg." and "Ext. length" are the percentages of exterior segments and exterior path length. Depth is the deepest sampled excursion; radius is the sampled covering radius. Distances are in normalised units; coverage and the coefficient of variation (CV) are defined in Appendix C.2. See Appendix C.5 for the comparison protocol.

Each outcome thus sits in the link the framework assigned it: the transitions between layers in the source, the containment failures in the geometry map. Decomposition brings back a fixed number of components, k, rather than one that grows with resolution. The first application kept the usual order of construction, geometry first, machine second. The second changes that order, because the next source of forced interruptions lies in the machine.

5 Second application: changing the construction order

The first application removed interruptions forced at the source. A connected source can still lead to interruptions at the machine. If obstacles divide the robot's safe postures into separate families, a tool path that passes between their regions forces posture resets, by the same counting argument. Which link fixes that count depends on the order in which the links are built. Every reviewed slicer constructs the tool path first. Geometry-first slicing fixes $x = \Phi \circ \xi$, which certifies the occupation measure through Φ, and then seeks a lift $q = \Psi_g[x]$ that follows it. Robot-aware methods do plan with machine constraints, but they take a geometric path or geometric elements as the input to that planning [9, 20, 21, 40], so obstacles, singularities, and posture changes are resolved for a geometry that is already fixed. Separating the geometry map from the lift exposes the order of construction as a design freedom of the chain. Reversing it is not a permutation of the same maps: Ψ_g lifts a prescribed tool path, whereas a machine-first construction needs a new map Ψ_k from the source into admissible configurations, and derives the tool path through forward kinematics (FK; Figure 6):

$$q = \Psi_k \circ \xi, \qquad x = \mathrm{FK} \circ q, \tag{10}$$

where Ψ_k takes values in the admissible configurations $\mathcal{Q}_{\mathrm{free}}$ and FK returns the tool position. Guarantees and obligations trade places. Static feasibility holds by construction, while coverage and the prescribed occupation measure, $x_{\#}\lambda^1 = \mu_\Omega$, which geometry-first obtains from Φ, become obligations. Configuration-first coverage exists in robotics: Yang et al. [8] plan coverage through configuration-space cells to minimise discontinuities. The framework adds the prescribed deposition measure and the link to source decomposition.

The density obligation. Let Ψ_k map into a regular submanifold $S \subseteq \mathcal{Q}_{\mathrm{free}}$, and let $f = \mathrm{FK}\,|_S$ be injective on a regular chart of S, with volume Jacobian J_S ($J_S = |\det Df|$ in equal dimension).



(a) Geometry first (b) Kinematics first

Figure 6: The two construction orders. (a) Geometry first: construct the tool path, then lift it to machine motion. (b) Kinematics first: construct admissible machine motion, then derive the tool path through forward kinematics; covering the target Ω with the prescribed measure remains an obligation. Vertices denote spaces and arrows indicate construction stages; dashed diagonals show the resulting composition. The lift Ψ_g acts on whole paths.

By change of variables, a configuration density a deposits the prescribed density exactly when

$$a(q) = \rho(f(q))\, J_S(q).$$

Uniform configuration density therefore need not give uniform deposition density. Where patches overlap, the target mass is partitioned first and each patch carries its share; multiple preimages must be summed rather than treated as one branch.

5.1 Posture sheets guide the source split

Once the order is reversed, the robot rather than the target can guide how the source is split, as the part's convexity guided it in the first application (Section 3.4). Call the path components of the safe fibre space $F = \{(x, q) \in \Omega \times \mathcal{Q}_{\text{free}} : \text{FK}(q) = x\}$ *sheets*, with workspace projections Ω_i; they are related to t-connectivity [41] and aspects [42]. A *posture switch* occurs where the one-sided lifted pairs $(x(t^{\pm}), q(t^{\pm}))$ lie in different sheets; a discontinuity within one sheet is not counted. A continuous stretch of a lifted path stays in one sheet, so a switch is the robot's analogue of a source transition. If a region A is reachable only in sheet 1 and a region B only in sheet 2, every passage of the tool path from A to B forces a switch, whatever lift is chosen.

The construction makes the split explicit. (1) Cover Ω by patches P_i, each inside the projection of one sheet. (2) Construct and validate a continuous safe lift $\sigma_i \colon P_i \to \mathcal{Q}_{\text{free}}$ in that sheet. (3) Partition the target mass among the patches, $w_i = \mu_\Omega(P_i)$, and give each patch its conditional density. (4) Traverse each patch with its own source piece, $q = \sigma_i \circ x_i$, and visit each patch once.

Proposition 5.1 (Sheets force switches; patches attain the minimum)**.** *Let* $\operatorname{supp} \mu_\Omega = \Omega$ *and let* c *be the fewest sheets whose projections cover* Ω*. (i) Every plan with* $x_\# \lambda^1 = \mu_\Omega$*, finitely many discontinuities, and safe one-sided limits makes at least* $c - 1$ *switches. (ii) If* Ω *is covered by* c *patches, each traversed with its share of* μ_Ω *by a continuous safe lift in one sheet, visiting each patch once attains* $c - 1$*. In particular, if two sheets are necessary and two such patches cover* Ω*, one switch is necessary and sufficient.*

Appendix D.1 gives the proof. Two distinctions bound the claim. Sheet coverage is not lift existence: a sheet whose projection covers a patch need not supply a continuous lift over it, so the lifts σ_i are hypotheses to validate. And a counted switch is an event in deposition time, whereas the physical repositioning between postures, with the flow off, is a separate collision-free motion that must be planned and checked. For a fixed path, the same argument fixes the count: with two sheets whose closed projections admit continuous sections, the minimum number of switches equals the number of alternations between $\Omega_1 \setminus \Omega_2$ and $\Omega_2 \setminus \Omega_1$, each switch being placed in the overlap. A geometry-first slicer inherits this count from its path; a better lift cannot lower it, just as a better order of source pieces cannot lower the number of transitions.

5.2 A three-link experiment

Setup. A planar arm with three revolute joints deposits on an annular sector among three disk obstacles (Figure 7(a)). Sampling the safe fibre gives two endpoint-anchored families, labelled Down and Up, whose projections overlap only for $\varphi \in [-4.4^\circ, 12.9^\circ]$, so neither covers Ω alone and $c = 2$ in the sampled model. Both pipelines use the same 4^m order-m Hilbert waypoints at orders $m = 2, 3, 4$, and two target densities: uniform, and a gradient density. *Geometry-first* is our baseline. It follows the Hilbert path over the whole chart, solves constrained inverse kinematics at each collocation point, and changes posture only when the current family becomes unavailable. *Kinematics-first* has a one-time preparation, which builds two patches, Down and Up, each with a stored, certified piecewise-affine joint-pose map, and a per-request step, which splits the source square at the target cut 4.25° into whole Hilbert rows per patch and reads joint poses from the stored map without any inverse-kinematics solve. The two groups are joined through a common handoff point in the overlap, where one posture reset occurs (Figure 7(b,d)). The *switch count* is the number of observed resets; the *coverage error* is the largest distance from a target grid to the sampled depositing path; the *density error* is the total-variation distance between target shares and waypoint shares on 4×8 area bins. Appendix D.2 gives the geometry, charts, preparation, certification, measurements, and timing protocol.

Results. Kinematics-first made one reset in all six requests and geometry-first made two, which equals the minimum for its fixed Hilbert path in the sampled availability model (Table 4). In that model, the bound $c - 1 = 1$ is attained in this example. The second geometry-first reset is a property of its path, not of its solver, and the reduction to one comes from the sheet-guided split of the source. Coverage errors are comparable and shrink with the order (at order four, 0.061 against 0.062 for uniform and 0.114 against 0.117 for gradient density). Density errors coincide for uniform density at every order. For the gradient density, they are slightly higher for kinematics-first (0.188 against 0.172 at order three and 0.062 against 0.043 at order four), consistent with the row rounding. Kinematics-first preparation took 14.2 s, after which each request took 5 to 39 ms to plan and certify, against 2.34 to 9.95 s for geometry-first. The speedup comes from reusing the prepared maps instead of solving inverse kinematics per request. Over the six requests in sequence, including preparation and the shared fibre analysis, kinematics-first's cumulative time falls below geometry-first's at the fourth request (22.4 against 22.7 s) and ends at 22.5 against 42.5 s.

The example completes the parallel with the first application: in both, the forced interruptions are counted by the path components at one link, and a source split matched to those components reaches the minimum: none for a connected volume on convex parts and one for two sheets.

		Resets		Coverage error		Density error		Planning (s)	
Density	m	G	K	G	K	G	K	G	K
Uniform	2	2	1	0.242	0.276	0.500	0.500	2.52	0.005
	3	2	1	0.121	0.139	0.000	0.000	4.92	0.013
	4	2	1	0.061	0.062	0.000	0.000	9.82	0.039
Gradient	2	2	1	0.398	0.363	0.457	0.469	2.34	0.005
	3	2	1	0.217	0.197	0.172	0.188	4.79	0.014
	4	2	1	0.114	0.117	0.043	0.062	9.95	0.039

Table 4: Three-link comparison of geometry-first (G) and kinematics-first (K) at 4^m waypoints. Resets are observed discrete posture changes. Coverage error is in target units; density error is total variation of original-waypoint shares on 4×8 area bins. Planning time is the median time to plan and certify one request. Kinematics-first also has a one-time preparation of 14.2 s; the shared fibre analysis (8.2 s) is excluded for both.

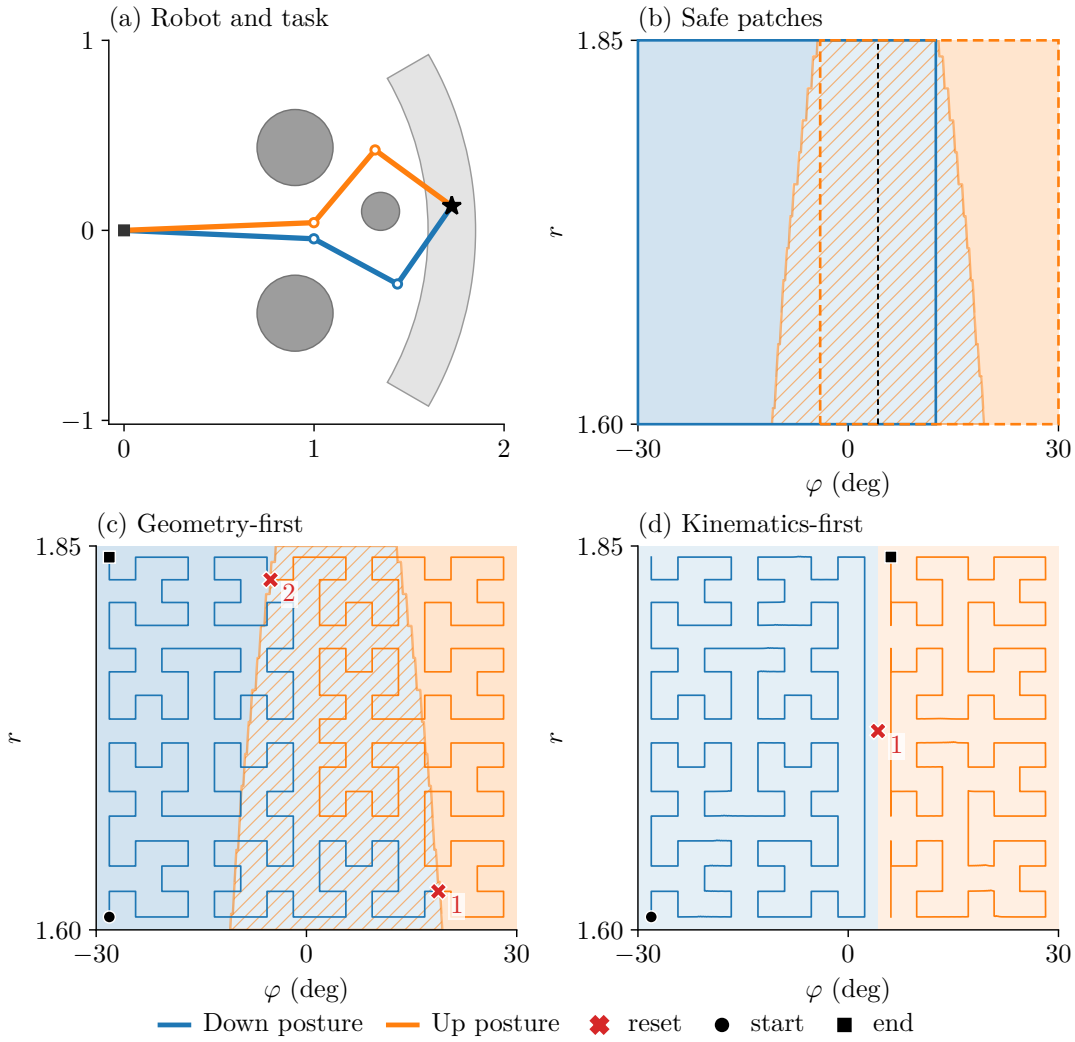


Figure 7: Three-link experiment at order four with uniform density. (a) Arm, obstacles, target sector, and the Down (blue) and Up (orange) postures at one tool point. (b) Robot-guided split: sampled availability of the two families over the chart, the prepared patch domains, and the assigned regions split at 4.25°. (c) Geometry-first path, shaded by sampled availability (one family or both), with two resets. (d) Kinematics-first path, shaded by assigned patch: patch 1 is completed, followed by one reset, then patch 2. Red crosses mark discrete, uncertified posture resets. The certified connectors to and from the handoff point are not drawn.

6 Outlook

The two applications demonstrate the framework; they do not exhaust it. Each was obtained by the same steps: read a slicer link by link, find a choice the field holds fixed, change it, and let the composition rules state what the new construction owes. We see three directions in which the same procedure extends.

Unexplored link choices. Table 1 is sparse. Every reviewed family builds geometry first and treats the process Θ as fixed, and the two constructions of this paper fill only two new cells. Other choices remain open. Sources can lie between a stack of layers and one volume, for instance a few connected pieces chosen to match the part and the machine at once. Traversals of a connected source need not be Hilbert curves, and other curves trade locality, smoothness, and turning differently. Construction orders can interleave the links rather than fix one first. A

process-aware order, for example, could let thermal or curing constraints in Θ shape the source split as the posture sheets did here. Each of these choices comes with its own obligations, and the composition rules say in advance what they are.

Lower bounds that combine several links. The two bounds of this paper each count components at one link. In a real slicer, obstructions arise at several links at once: a part decomposed into k near-convex pieces, printed by a robot whose safe postures form c sheets. Treated separately, they give at least $k-1$ transitions and $c-1$ switches. A plan with no extra interruptions would have to align the source pieces with the sheets. When they do not align, we expect the true minimum to be counted by the components of a combined space, consisting of pairs of source pieces and sheets that a continuous stretch can occupy together, and to exceed both separate bounds. Stating and proving such bounds, and building sources that attain them, would turn the counting argument from a diagnosis of one link into a design rule for the whole chain.

Uses beyond deposition. None of these steps is specific to deposition. Any process that covers a region with a prescribed measure along a path executed by a machine has the same chain. Examples are machining, where the measure is material removed; inspection, where it is sensor dwell; and spraying and welding, where it is coating or heat input. Each has its own forced interruptions, such as tool retractions, re-fixturing, and torch restarts, and its own machine with posture sheets. We expect the remaining unexplored link choices and orders, and lower bounds that combine obstructions from several links, to be the next uses of the framework, within slicing and beyond it.

7 Conclusion

Forced interruptions in slicing are topological (Section 1): the map chain shows which link supplies the components and what a construction that changes that link, or the order of construction, must guarantee in its place.

The two applications find these components at two different links. In the first, they are the pieces of the source. A connected volume as the source removes them and leaves two obligations: a continuous measure-preserving traversal and a continuous geometry map. A Hilbert curve and optimal transport meet these obligations on convex parts. On non-convex parts, decomposing the source repairs the remaining failures up to each piece's hull gap, at the cost of a fixed number of transitions. In the second application, the components are the robot's posture sheets. Building admissible machine motion first lets the sheets guide the source split, and patches with continuous safe lifts, each visited once, attain the minimum.

Neither algorithm alone shows that stops and posture switches obey one counting argument, attained in both cases by a source split matched to the components; the framework does. Maps that stay inside non-convex parts, sequencing of decomposed pieces, and smoothing that preserves containment and respects machine dynamics are the natural next steps, alongside the directions of Section 6. Slicing has been developed by improving paths within a fixed chain. The chain itself—its links, their order, and the components each one supplies—is where the forced limits of a slicer are set, and where new slicers can be designed.

A Framework and interruption guarantees

This appendix supports the framework claims of Section 2: that a slicer is a chain of maps, that guarantees compose from link to link, and that a disconnected source forces transitions. Subsection A.1 fixes the spaces, the deposition-time convention, and the formal definition of a geometry-first map chain. Subsection A.2 states and proves the composition rules. Subsection A.3 proves the transition bound for disconnected sources and its propagation into tool-path, machine, and deposition interruptions.

A.1 Spaces, paths, and the map chain

Let $\Omega \subset \mathbb{R}^3$ be compact, with target deposition measure μ_Ω. All measures are probability measures, and $f_\# \eta(A) = \eta(f^{-1}(A))$ is the pushforward of η by a measurable map f. The target measure may be volumetric or live on lower-dimensional sets such as layers. Let $\mathcal{Q}$ be the configuration space of the machine [43], with the tool-position forward-kinematics map

$$\mathrm{FK} : \mathcal{Q} \to \mathbb{R}^3, \qquad x = \mathrm{FK} \circ q. \tag{11}$$

The static admissible set $\mathcal{Q}_{\text{free}} \subseteq \mathcal{Q}$ encodes joint limits, a prescribed singularity margin, and avoidance of fixed obstacles.

Deposition time. Time $t \in I = [0, 1]$ is normalised deposition time, and λ^1 is the uniform probability measure on I. For a space M, $\mathrm{PC}(M)$ denotes the paths $I \to M$ that are continuous on the pieces of a finite partition of I, with one-sided limits at the breakpoints; $C^0(I, M) \subseteq \mathrm{PC}(M)$. Physical execution crosses each discontinuity with a non-deposition travel move, which takes machine time but no deposition time. The traversal ξ, tool path x, and joint path q share the time parameter.

Definition A.1 (Map chain)**.** For a prescribed target (Ω, μ_Ω), a geometry-first map chain is a tuple $(\mathcal{X}, \mu_\mathcal{X}, \xi, \Phi, \Psi_g)$, where:

- $\mathcal{X}$ is a compact manifold with boundary, or a finite disjoint union of such manifolds, of dimension at most three;
- $\mu_\mathcal{X}$ is a probability measure on $\mathcal{X}$;
- $\xi \in \mathrm{PC}(\mathcal{X})$ is the traversal, satisfying $\xi_\# \lambda^1 = \mu_\mathcal{X}$;
- $\Phi : \mathcal{X} \to \Omega$ is a measurable geometry map, satisfying $\Phi_\# \mu_\mathcal{X} = \mu_\Omega$;
- $\Psi_g : \mathcal{D} \to \mathrm{PC}(\mathcal{Q})$ is a configuration lift on a declared domain of paths $\mathcal{D} \subseteq \mathrm{PC}(\Omega)$, satisfying

$$\mathrm{FK} \circ \Psi_g[x] = x \qquad \text{for all } x \in \mathcal{D}. \tag{12}$$

The lift applies only when x has finitely many discontinuities, which is not automatic for a measurable Φ (Subsection B.2); applicability of the lift and static admissibility therefore require separate certificates. With both pushforward conditions, (2) gives the tool path the occupation measure μ_Ω. The process operator Θ, which maps a joint path to a manufactured set for a fixed process and protocol, is left abstract.

Definition A.1 and the certificates below describe the geometry-first order, in which the tool path is built and then lifted. The kinematics-first construction of Section 5 is not a permutation of these maps: it introduces a new map Ψ_k from the source into $\mathcal{Q}_{\text{free}}$ and derives the tool path through forward kinematics (10), so static feasibility becomes a certificate of Ψ_k, while the occupation measure becomes the density obligation stated there.

The lift acts on whole paths because, for an n-joint arm at a regular value of FK, the fibre $\mathrm{FK}^{-1}(x)$ is locally $(n-3)$-dimensional [44]: no configuration in it is canonical, and singularities can prevent a globally continuous inverse branch [9, 45, 46]. A lift is therefore selected along the whole path [8, 20, 47].

The planar chain. In the planar chain, D_k is the cross-section at height z_k, and the label k keeps the slices apart in the source even though the solid is connected. The ideal source measure is $\mu_{\mathcal{X}} = \sum_k w_k \bar{\lambda}_{D_k}$, with $w_k \geq 0$ the time share of layer k and $\bar{\lambda}_{D_k}$ the normalised area measure on $D_k \times \{k\}$. A finite zigzag only approximates this area measure; taking $\mu_{\mathcal{X}}$ to be the normalised occupation measure of the zigzag centreline instead gives an exact chain for the finite raster.

A.2 Certificates that compose

A *certificate* is a property of one stage alone, on a specified domain. Five kinds are used:

- *Coverage:* a pushforward identity, or a measure error bounded in W_∞.
- *Regularity:* a modulus of continuity $\omega_f(\delta) = \sup_{d(a,a')\leq\delta} d(f(a), f(a'))$.
- *Discontinuity:* the set $\Sigma(f)$ of discontinuity points, or a certified set containing it.
- *Containment:* a region containing the image of the stage.
- *Stability:* a uniform implementation error $\|\hat{f} - f\|_\infty \leq \varepsilon_f$, measured in the target space of the stage.

For probability measures on a compact metric space, W_∞ [48] is the smallest r such that some coupling moves no mass farther than r. For paths with discontinuities, regularity certificates hold on each continuous piece.

Proposition A.2 (Composition rules)**.** *Let $f\colon A \to B$ and $g\colon B \to C$ be measurable stages between compact metric spaces and μ a probability measure on A, with ν and κ target probability measures on B and C.*

(i) Coverage. $W_\infty((g \circ f)_\# \mu, \kappa) \leq \omega_g(W_\infty(f_\# \mu, \nu)) + W_\infty(g_\# \nu, \kappa)$. *In particular, exact pushforward certificates compose exactly.*

(ii) Regularity. $\omega_{g\circ f} \leq \omega_g \circ \omega_f$.

(iii) Discontinuity. $\Sigma(g \circ f) \subseteq \Sigma(f) \cup f^{-1}(\Sigma(g))$.

(iv) Containment. *If $f(A) \subseteq B'$ and $g(B') \subseteq C'$, then $(g \circ f)(A) \subseteq C'$.*

(v) Stability. *If implementations $\hat{f}, \hat{g}$ satisfy $\|\hat{f} - f\|_\infty \leq \varepsilon_f$ on A and $\|\hat{g} - g\|_\infty \leq \varepsilon_g$ on $\hat{f}(A)$, with ω_g taken on $f(A) \cup \hat{f}(A)$, then $\|\hat{g} \circ \hat{f} - g \circ f\|_\infty \leq \varepsilon_g + \omega_g(\varepsilon_f)$.*

Proof. For (i), an optimal W_∞ coupling of $f_\#\mu$ and ν exists on compact spaces [48]; push it forward by $g \times g$ and apply the triangle inequality for W_∞. Items (ii)–(iv) are immediate from the definitions, and (v) follows from $\|\hat{g} \circ \hat{f} - g \circ \hat{f}\|_\infty + \|g \circ \hat{f} - g \circ f\|_\infty$. □

Rule (iii) is an upper bound; Proposition A.5 gives the matching lower bound for disconnected sources. Rule (ii) multiplies Hölder exponents. Repairing one stage does not repair another: denser sampling cannot fix a wrong geometry map, and a better lift cannot fill missing coverage.

Corollary A.3 (W_∞ gives a covering radius)**.** *If $W_\infty(\nu, \mu_\Omega) \leq r$, then every point of* $\operatorname{supp}\mu_\Omega$ *lies within distance r of* $\operatorname{supp}\nu$.

Proof. An optimal coupling matches almost every target point to a point of $\operatorname{supp}\nu$ within distance r. Distance to this closed set is continuous, and every neighbourhood of a point of $\operatorname{supp}\mu_\Omega$ has positive mass, so the bound extends to all of $\operatorname{supp}\mu_\Omega$. □

With $\operatorname{supp}\mu_\Omega = \Omega$, one coupling bound therefore controls both the distribution of deposition time and the covering radius of the path.

Because Ψ_g acts on paths, its certificates take a different form: a nondecreasing transfer function T_Ψ, such that an input path with modulus at most ω lifts to a joint path with modulus at most $T_\Psi(\omega)$; a set B_Ψ of branch-switch times, with $\Sigma(q) \subseteq \Sigma(x) \cup B_\Psi$; and a region $U \subseteq \Omega$ whose admitted paths lift into $\mathcal{Q}_{\text{free}}$. A pointwise section with Lipschitz constant L_Ψ has $T_\Psi(\omega) = L_\Psi\, \omega$. Table 5 assembles the certificates; replacing a stage changes only its column.

Property	ξ	Φ	Ψ_g	Resulting guarantee
Coverage	$\xi_{\#}\lambda^1 = \mu_{\mathcal{X}}$	$\Phi_{\#}\mu_{\mathcal{X}} = \mu_\Omega$	$\mathrm{FK} \circ \Psi_g[x] = x$	$x_{\#}\lambda^1 = \mu_\Omega$
Regularity	ω_ξ	ω_Φ	T_Ψ	$\omega_q \le T_\Psi(\omega_\Phi \circ \omega_\xi)$
Discontinuity	$\Sigma(\xi)$	$\Sigma(\Phi)$	B_Ψ	$\Sigma(q) \subseteq \Sigma(\xi) \cup \xi^{-1}(\Sigma(\Phi)) \cup B_\Psi$
Static feasibility	–	$\Phi(\mathcal{X}) \subseteq U$	$\mathrm{PC}(U) \to \mathrm{PC}(\mathcal{Q}_{\mathrm{free}})$	$q \in \mathrm{PC}(\mathcal{Q}_{\mathrm{free}})$

Table 5: Certificates along the geometry-first chain. Each resulting guarantee is a property of the tool path x or joint path q, or of the occupation measure of x, and depends only on the certificates in its own row; none describes a manufactured part. Feasibility is with respect to the static set $\mathcal{Q}_{\mathrm{free}}$ only.

A.3 Transitions of disconnected sources

Suppose the source is a finite disjoint union of compact connected components,

$$\mathcal{X} = \bigsqcup_{k=1}^{N} \mathcal{X}_k. \tag{13}$$

Planar layers give at least one component per layer; curved layers [10, 19], separate surfaces [22], and parameterisation patches [4] give others. The components are those of the source, not of the part: the images $\Phi(\mathcal{X}_k)$ can touch or overlap in Ω. Four kinds of interruption must be kept apart.

Definition A.4 (Interruptions)**.**

- A *source interruption* is a discontinuity of the traversal ξ; a *transition* is a source interruption at which ξ passes from one component to another.
- A *tool-path interruption* is a discontinuity of $x = \Phi \circ \xi$.
- A *machine interruption* is a discontinuity of the joint path $q = \Psi_g[x]$.
- A *deposition interruption* is a stop of the material flow during physical execution.

The first three occur in deposition time; the last occurs in machine time, which also includes travel moves.

Proposition A.5 (Source-component transitions)**.** *Let $\mathcal{X}$ be as in* (13)*, let $\xi \in \mathrm{PC}(\mathcal{X})$ satisfy $\xi_{\#}\lambda^1 = \mu_{\mathcal{X}}$, and suppose $\mu_{\mathcal{X}}(\mathcal{X}_k) > 0$ for every k.*

(i) The traversal ξ has at least $N-1$ transitions, whatever the order in which it visits the components.

(ii) Suppose Φ is continuous on each component. At a transition t_i, the tool path $x = \Phi \circ \xi$ has the one-sided limits $x(t_i^{\pm}) = \Phi(\xi(t_i^{\pm}))$, and it is discontinuous at t_i unless

$$\Phi(\xi(t_i^-)) = \Phi(\xi(t_i^+)). \tag{14}$$

Proof. (i) Let $0 = t_0 < \cdots < t_m = 1$ be a partition on whose open pieces ξ is continuous. The image of each piece is connected, so it lies in one component; label the piece by that component. Each component has positive mass, so by $\xi_{\#}\lambda^1 = \mu_{\mathcal{X}}$ it is visited for a positive time, and its label appears on at least one piece. Merge consecutive pieces with the same label. The resulting sequence contains all N labels, with adjacent labels distinct, so it changes label at least $N-1$ times, and each change is a transition. (ii) $\xi(t) \to \xi(t_i^-)$ within one compact component as $t \to t_i^-$, and Φ is continuous there; the same holds from the right. □

Condition (14) can hold when two components touch in Ω, as adjacent patches do.

Corollary A.6 (Separated images give tool-path interruptions)**.** *Under the hypotheses of Proposition A.5(ii), suppose that $\mathrm{dist}(\Phi(\mathcal{X}_k), \Phi(\mathcal{X}_\ell)) \ge \Delta > 0$ for all $k \ne \ell$. Then every transition is a tool-path interruption with $\|x(t_i^+) - x(t_i^-)\| \ge \Delta$, and x has at least $N-1$ such interruptions.*

Proof. At a transition from $\mathcal{X}_k$ to $\mathcal{X}_\ell$, the limits $x(t_i^-) \in \Phi(\mathcal{X}_k)$ and $x(t_i^+) \in \Phi(\mathcal{X}_\ell)$ are at least Δ apart. Apply Proposition A.5(i). □

Since FK is continuous, q cannot be continuous where x is not, so every tool-path interruption is a machine interruption. By the deposition-time convention of Subsection A.1, it is also a deposition interruption, crossed by a travel move with the flow off. The converse implications fail: a connected source can still have a discontinuous traversal or geometry map, and a continuous tool path can still produce machine interruptions through a branch switch or a lift that fails near a singularity [8, 9]. A connected source therefore removes only the lower bound of Proposition A.5; continuity of the later maps must be certified separately, as Appendix B does for the volumetric application.

For planar layers at spacing h with all weights $w_k > 0$, $\Phi(u, v, k) = (u, v, z_k)$ is continuous on each component and distinct layers lie at least h apart, so Corollary A.6 with $\Delta = h$ is the formal version of the stop count in Section 2.3.

B Proofs for the volumetric application

This appendix proves the guarantees that Section 3 states for the first application. Subsection B.1 establishes the Hilbert traversal's measure preservation, locality, and finite-order estimates. Subsection B.2 gives transport existence and the assumptions under which the map is continuous on the closed cube. Subsection B.3 derives the ideal-path guarantees and the conditional finite-waypoint bounds. Subsection B.4 proves what the computed centroid path guarantees, including the hull-gap bounds used by decomposition. Two objects are kept apart throughout: the ideal transport map Φ, and the computed centroids $\hat{x}_i$, whose equal cell masses do not imply pointwise closeness to Φ. All results concern the tool path and its occupation measure.

B.1 Source and traversal

The transported path is the geometry-first chain of Definition A.1 with

$$\mathcal{X} = [0,1]^3, \qquad \mu_{\mathcal{X}} = \lambda^3, \qquad \xi = \mathcal{H}_3, \tag{15}$$

where λ^3 is the uniform probability measure on the cube and Φ is the optimal-transport map defined below. Any lift Ψ_g completes the tuple; the prototype stops at the tool path. The cube is connected, so Proposition A.5 forces no transition; tool-path interruptions can still come from $\Sigma(\Phi)$, and machine interruptions from B_Ψ.

For each order $m \geq 1$, split the cube into 8^m dyadic cells of side 2^{-m} and the time interval into 8^m blocks of length 8^{-m}. The recursive orders must be compatible: at each level, every cell is listed once, consecutive cells share a face, and the eight children of each cell are listed consecutively at the next level. Such constructions exist [28], though not every three-dimensional Hilbert variant has them. The curve $\mathcal{H}_3$ [25–27] has the following properties.

(H1) *Continuity and surjectivity.* $\mathcal{H}_3$ is continuous and passes through every point of the cube.
(H2) *Measure preservation.* $\mathcal{H}_{3\#}\lambda^1 = \lambda^3$.
(H3) *Hölder continuity.* $\|\mathcal{H}_3(t) - \mathcal{H}_3(t')\| \leq 2\sqrt{6}\,|t - t'|^{1/3}$.

Continuity and surjectivity are standard for compatible recursive Hilbert traversals [26–28]; measure preservation for the self-similar construction is stated in [28, Section 2.4]. For the selected nested cell orders, it also follows directly: the assigned cell-centre maps H_m converge uniformly to $\mathcal{H}_3$, and for every continuous F,

$$\int_I F(\mathcal{H}_3(t))\,dt = \lim_m 8^{-m} \sum_i F(u_i) = \int_{[0,1]^3} F(u)\,du.$$

The explicit constant in (H3) uses only face adjacency: if $8^{-(m+1)} \le |t - t'| < 8^{-m}$, the two points lie in the same or consecutive level-m cells, whose union has diameter at most $\sqrt{6}\,2^{-m} \le 2\sqrt{6}\,|t - t'|^{1/3}$.

The order-m curve is the sequence of cell centres $u_1, \dots, u_{8^m}$; face adjacency gives $\|u_{i+1} - u_i\| = 2^{-m}$. Sending the mass of each cell to its centre moves no mass farther than half a cell diagonal, so the empirical measure $\nu_m = 8^{-m} \sum_i \delta_{u_i}$ satisfies

$$W_\infty(\nu_m, \lambda^3) \le \tfrac{\sqrt{3}}{2}\, 2^{-m}. \tag{16}$$

These finite estimates need no transport regularity.

B.2 Transport existence and regularity

The geometry map is the solution of the Monge problem with quadratic cost,

$$\Phi = \operatorname*{arg\,min}_{T:\, T_\# \lambda^3 = \mu_\Omega} \int_{[0,1]^3} \|u - T(u)\|^2 \, du. \tag{17}$$

Existence almost everywhere. Since λ^3 has a density and both measures have compact support, Brenier's theorem gives a solution, unique λ^3-almost everywhere, of the form $\Phi = \nabla\varphi$ with φ convex [29, 30]. This gives existence and uniqueness, not continuity. Without convexity, smooth densities still give a map that is smooth outside a closed singular set of measure zero [49]. If Φ were discontinuous across a surface inside the cube, the Hilbert curve would cross it in every cell it cuts; with distinct one-sided limits at infinitely many crossings, x would have infinitely many discontinuities and the lift of Definition A.1 would not apply. This is why a finite discontinuity set is a certificate, and why convexity matters. For the torus, the L-bracket, and the complex part, only this almost-everywhere map is available; neither non-convexity nor the jumps measured in Appendix C proves a singular set.

Continuity on the closed cube. Suppose Ω is compact and convex with nonempty interior, and $\mu_\Omega = \rho\, \mathcal{L}^3|_\Omega$ with $0 < a \le \rho \le b < \infty$ almost everywhere. Both densities are bounded above and below by positive constants on convex domains, so they define doubling measures, and the global regularity theory of Jhaveri and Savin supplies a Hölder-continuous representative $\Phi = \nabla\varphi$ on $[0,1]^3$ for some $0 < \beta < 1$ [34, Theorem 1.1, Corollary 2.5, and Lemma 2.6], extending Caffarelli's interior theory [33]. Changing an almost-everywhere representative preserves the pushforward, and smooth or uniformly convex boundaries are not required. This covers uniform measures on full-dimensional convex boxes, cylinders, spheres, and convex polyhedra. It supplies no numerical value of the Hölder constant, and convexity alone is insufficient when the target density vanishes, is unbounded, or lives on a lower-dimensional set.

B.3 Path certificates

Convex regularity gives the geometry stage the certificate

$$\|\Phi(u) - \Phi(u')\| \le C_\Phi \|u - u'\|^\beta \qquad \text{for all } u, u' \in [0,1]^3, \tag{18}$$

with some $\beta \in (0, 1]$ and finite but unknown C_Φ. For a non-convex part it remains a hypothesis. An implementation produces waypoints $\hat{x}_i$ with error $\varepsilon = \max_i \|\hat{x}_i - \Phi(u_i)\|$.

Theorem B.1 (Ideal-path and finite-waypoint guarantees)**.** *(i)* Conditional bounds. *Under* (18)*, the ideal path $x = \Phi \circ \mathcal{H}_3$ is continuous, $(\beta/3)$-Hölder, and has occupation measure $x_\# \lambda^1 = \mu_\Omega$. Waypoints within ε of $\Phi(u_i)$ satisfy*

$$W_\infty\Big(8^{-m} \textstyle\sum_i \delta_{\hat{x}_i},\, \mu_\Omega\Big) \le C_\Phi \big(\tfrac{\sqrt{3}}{2}\big)^\beta 2^{-\beta m} + \varepsilon, \qquad \|\hat{x}_{i+1} - \hat{x}_i\| \le C_\Phi 2^{-\beta m} + 2\varepsilon. \tag{19}$$

The first bound also bounds the target covering radius.

(ii) Convex setting. *If Ω is compact and convex with nonempty interior and its target density is bounded above and below by positive constants, then* (18) *holds for some $\beta \in (0,1]$ and finite C_Φ. Thus (i) applies: the ideal path has no tool-path interruption and exactly the prescribed occupation measure.*

Proof. For (i), (H3) and Proposition A.2(ii) give Hölder continuity, and (H2) gives $x_\# \lambda^1 = \Phi_\# \lambda^3 = \mu_\Omega$. Apply Proposition A.2(i) to ν_m and Φ, using (16); moving each waypoint by at most ε adds ε. Corollary A.3 supplies the covering-radius bound. Face adjacency gives $\|u_{i+1} - u_i\| = 2^{-m}$, so (18) bounds the ideal step by $C_\Phi 2^{-\beta m}$, and the triangle inequality adds 2ε. For (ii), the closed-domain regularity of Subsection B.2 supplies (18). □

A lift with Lipschitz constant L_Ψ on a region containing the waypoints bounds the joint steps by $L_\Psi(C_\Phi 2^{-\beta m} + 2\varepsilon)$.

The finite bounds are conditional on ε, which is not certified for the computed path: the semi-discrete computation below certifies equal cell mass and the measured bound D_m instead, and neither bounds ε.

B.4 Computed-path guarantees

Power cells and equal-mass optimality. Let $\mu_\Omega = \rho\, dy$ with $\rho > 0$ integrable on Ω, let $u_1, \dots, u_n$ be the distinct sites of the order-m curve, and let $\nu_m = n^{-1} \sum_i \delta_{u_i}$. For weights ψ, the power cell $P_i(\psi) = \{y \in \mathbb{R}^3 : \|y - u_i\|^2 - \psi_i \le \|y - u_j\|^2 - \psi_j \ \forall j\}$ is a convex polyhedron, and $L_i = P_i \cap \Omega$ is the Laguerre cell (7). Two cells meet only where two power distances are equal, on a plane, so the cells tile Ω up to a null set. Weights satisfying (8) make the map $y \mapsto u_i$ on L_i the optimal transport from μ_Ω to ν_m [35–37]. The waypoints are the centroids $\hat{x}_i$ of (9), and $D_m = \max_i \operatorname{diam} L_i$.

Proposition B.2 (What equal cell mass controls)**.** *Let ψ satisfy* (8) *and $\hat{\nu}_m = n^{-1} \sum_i \delta_{\hat{x}_i}$.*

(i) Distribution. *$W_\infty(\hat{\nu}_m, \mu_\Omega) \le D_m$.*

(ii) Order. *$\hat{x}_i \in P_i(\psi)$, so $\hat{x}_i$ is power-nearest to its own site u_i.*

(iii) Waypoint density. *For every Borel set A, with $A^+ = \{y : \operatorname{dist}(y, A) \le D_m\}$ and $A^- = \{y : \operatorname{dist}(y, \mathbb{R}^3 \setminus A) > D_m\}$,*

$$\mu_\Omega(A^-) \le \hat{\nu}_m(A) \le \mu_\Omega(A^+), \qquad \left|\hat{\nu}_m(A) - \mu_\Omega(A)\right| \le \mu_\Omega(A^+ \setminus A^-).$$

(iv) Containment. *If Ω is convex, every $\hat{x}_i$ and every segment $[\hat{x}_i, \hat{x}_{i+1}]$ lies in Ω.*

(v) Consistency. *If $\Omega = [0,1]^3$ and ρ is constant, $\psi = 0$ solves* (8)*, each L_i is the Hilbert cell of u_i, and $\hat{x}_i = u_i$.*

Proof. $\hat{x}_i$ is the mean of the probability measure $n\, \mu_\Omega|_{L_i}$, so it lies in the closed convex hull of L_i. For $y \in L_i$, the distance from y to a point of that hull is at most the largest distance from y to a point of $\overline{L_i}$, hence at most $\operatorname{diam} L_i \le D_m$. (i) The coupling $\gamma = \sum_i \mu_\Omega|_{L_i} \otimes \delta_{\hat{x}_i}$ has first marginal μ_Ω, because the cells tile Ω up to a null set, and second marginal $\sum_i \mu_\Omega(L_i)\delta_{\hat{x}_i} = \hat{\nu}_m$ by (8); it moves no mass farther than D_m. (ii) $L_i \subseteq P_i$, which is closed and convex, so it contains the hull of L_i and hence $\hat{x}_i$. (iii) If $\hat{x}_i \in A$, every point of L_i is within D_m of $\hat{x}_i$, so $L_i \subseteq A^+$; summing over these disjoint cells gives $\hat{\nu}_m(A) \le \mu_\Omega(A^+)$. If L_i meets A^- at some y, then $\|\hat{x}_i - y\| \le D_m < \operatorname{dist}(y, \mathbb{R}^3 \setminus A)$, so $\hat{x}_i \in A$; the cells meeting A^- cover it, which gives $\mu_\Omega(A^-) \le \hat{\nu}_m(A)$. Since $A^- \subseteq A \subseteq A^+$, $\mu_\Omega(A)$ lies in the same interval, and the difference is at most its length. (iv) $L_i = P_i \cap \Omega$ is convex and contains $\hat{x}_i$; a convex Ω contains the segment between two of its points. (v) With $\psi = 0$, the power cells are the Voronoi cells of the cube grid, that is, the Hilbert cells, each of volume $1/n$ and centroid u_i. □

A computed solution meets (8) only to a relative tolerance τ, and (i) and (iii) then hold up to τ. Item (iii) makes the waypoint share of a region accurate only when the region is thick

compared with D_m. Nothing in the proposition bounds D_m or keeps $\hat{x}_i$ in a non-convex Ω: a clipped cell can be non-convex or disconnected, its centroid can lie outside, and its diameter can stay large as m grows. Nor does it bound $\varepsilon = \max_i \|\hat{x}_i - \Phi(u_i)\|$: on convex parts quantitative stability results give L^2 rates [50], but no pointwise rate is established here.

Convex-hull gap. For a nonempty compact domain $K \subset \mathbb{R}^3$, define its *convex-hull gap* by

$$\delta_K = \sup_{z \in \mathrm{conv}(K)} \mathrm{dist}(z, K). \tag{20}$$

Near-convexity means that δ_K is small at the geometric scale of interest. The domain K may be the whole part or any compact subdomain.

Proposition B.3 (Convex-hull gap bounds centroid-path excursions)**.** *Let $K \subset \mathbb{R}^3$ be nonempty and compact, let $\rho \geq 0$ be integrable on K, and let $L_1, \ldots, L_n \subseteq K$ be measurable cells with positive masses. Define*

$$\hat{x}_i = \frac{\int_{L_i} y\, \rho(y)\, dy}{\int_{L_i} \rho(y)\, dy}.$$

Every $\hat{x}_i$ and every straight segment $[\hat{x}_i, \hat{x}_\ell]$ lie in $\mathrm{conv}(K)$ and have distance at most δ_K from K. In particular, for any polyline Γ whose vertices are chosen from these centroids,

$$\sup_{z \in \Gamma} \mathrm{dist}(z, K) \leq \delta_K, \qquad \Gamma \subseteq K \oplus \bar{B}_{\delta_K}. \tag{21}$$

If $K \subseteq \Omega$, the same bound holds for distance to Ω. Moreover, $\delta_K = 0$ if and only if K is convex, in which case containment in K is exact.

Proof. The convex hull $H = \mathrm{conv}(K)$ of a compact set in finite-dimensional space is compact, hence closed and convex. Each $\hat{x}_i$ is the mean of the normalised measure with density ρ on L_i, whose support lies in K. Such a mean belongs to its closed convex hull: otherwise a separating hyperplane would put the mean strictly beyond all points of its support, contradicting the defining integral. Thus $\hat{x}_i \in H$. Convexity of H contains every segment $[\hat{x}_i, \hat{x}_\ell]$ and therefore Γ. Equation (20) bounds the distance to K of every point in H, proving (21). If $K \subseteq \Omega$, then $\mathrm{dist}(z, \Omega) \leq \mathrm{dist}(z, K)$. Finally, convexity gives $H = K$ and $\delta_K = 0$. Conversely, if $\delta_K = 0$, every point of H has zero distance to the closed set K, so $H \subseteq K$ and $H = K$. □

The bound does not require equal cell masses, convex cells, a particular visiting order, or regularity of the transport map, and it bounds excursion depth only. A volume ratio $|K|/|\mathrm{conv}(K)|$ cannot replace δ_K, because a narrow indentation may have little missing volume and substantial depth.

Partitioned parts.

Corollary B.4 (Independent transport on partitioned parts)**.** *Let $\Omega = \bigcup_{j=1}^{k} \Omega_j$ be a union of compact pieces with $\mu_\Omega(\Omega_j \cap \Omega_\ell) = 0$ for $j \neq \ell$ and positive target-mass shares $w_j = \mu_\Omega(\Omega_j)$. For $\mu_\Omega = \rho\, dy$, run the algorithm independently on each piece using its conditional target measure*

$$\mu_j = w_j^{-1} \mu_\Omega|_{\Omega_j},$$

whose density on Ω_j is ρ/w_j. Give piece j deposition-time share w_j. For an equal-mass solve with n_j Hilbert sites, each local cell has target mass $1/n_j$ under μ_j, corresponding to global mass and waypoint time share w_j/n_j. Let Γ_j be its Hilbert-ordered centroid polyline and δ_j the hull gap (20) of Ω_j. Then

$$\sup_{z \in \Gamma_j} \mathrm{dist}(z, \Omega_j) \leq \delta_j.$$

If Ω_j is convex, $\delta_j = 0$ and $\Gamma_j \subseteq \Omega_j$.

Proof. Apply Proposition B.3 with $K = \Omega_j$ and density ρ/w_j to each local solve. The mass allocation follows from $\mu_\Omega = \sum_j w_j \mu_j$ and $\mu_\Omega(L_{ji}) = w_j \mu_j(L_{ji}) = w_j/n_j$ for each local cell L_{ji}. □

C Volumetric experimental methods and supporting results

This appendix supplies the methods and the additional results behind the evaluation of the first application in Section 4. Subsections C.1 and C.2 give the implementation, metrics, and baselines. Subsection C.3 reports all fifteen whole-part runs. Subsection C.4 separates the two non-convex failure mechanisms. Subsection C.5 documents the partitions, hull-gap evaluation, and same-budget comparison. The methods of the kinematics-first experiment are in Appendix D.

C.1 Implementation

Geometry. A watertight polygonal solid is scaled into $[0, 1]^3$ with a margin of 0.05, and all distances are in these normalised units. The solid is divided into tetrahedra by TetGen [39] without changing its surface. The sites are the $n = 8^m$ cell centres of the order-m Hilbert curve, generated with Skilling's integer transform [51]; for orders 1 to 6 we checked the face-adjacent, nested ordering that Subsection B.1 assumes. No waypoint is projected, removed, or smoothed.

Solver. The equal-mass weights (8) are computed by the unmodified semi-discrete solver of Geogram 1.10.1 [37]. The solve runs coarse to fine: at order two, initial weights are chosen so that every cell has positive mass, and Newton iterations [38] follow; each higher order starts from the previous order's weights, interpolated onto the nested grid. The stopping threshold bounds each relative cell-mass error by 10^{-6}. The centroids (9) are integrated exactly over the clipped cells with a degree-two tetrahedral rule.

Acceptance. A result is accepted only if every relative cell-mass error is below 10^{-6}, no cell is empty, every centroid is finite, and every centroid is power-nearest to its own site, which confirms that the waypoints keep their Hilbert labels (Proposition B.2(ii)). Every transport solve reported here passed; acceptance does not certify geometric containment. On the unit cube, the solver returned zero weights and centroids equal to the sites to 10^{-15}, as Proposition B.2(v) requires.

Evaluation domain. In the whole-part series, containment and segment metrics for the cylinder, sphere, and torus use the analytic solids; the decomposition experiments use the polygonal meshes.

C.2 Metrics and baselines

Cloud metrics ignore the order of the waypoints. Containment is the fraction of waypoints inside Ω. Coverage is the fraction of voxels of a 20^3 grid that meet Ω and contain a waypoint, and its coefficient of variation (CV) is that of the waypoint counts over those voxels, lower being more uniform. The cell diameter D_m is the largest diameter of a clipped Laguerre cell. *Sequence metrics* depend on the order. Jumps count steps longer than five times the mean step; this is a resolution-dependent indicator, not a continuity test. *Segment metrics* sample each segment at a spacing of at most 10^{-3}; the excursion depth is the distance from an exterior sample to the surface. The covering radius is the largest distance from a point of the evaluation sample to its nearest waypoint.

The parts are a cylinder, a sphere, a solid torus, an L-bracket with a reentrant edge, and a complex part consisting of a block with several holes and a boss (genus three). The evaluation

sample is a fixed, seeded set of 10 000 uniform interior points; the map itself involves no random choice. An independent uniform sample of the same size provides a coverage reference.

The layered baseline visits the cells of the grid of side 2^{-m} whose centres lie in Ω, layer by layer, traversing each layer row by row in alternating directions; it is the planar slicer with $h = 2^{-m}$. Every move between cells that do not share a face is a travel move, hence a stop, and moves between layers are counted separately. For the transported path, each maximal *exterior run* of segments leaving Ω counts as one stop.

C.3 Whole-part runs

Setup. Each of the five parts is solved as a single target with uniform density at orders four, five, and six, giving fifteen runs.

Containment and coverage. Table 6 reports every uniform run. On the torus at order four, coverage is 0.882 with CV 0.613, against 0.861 and 0.732 for the uniform reference. Re-evaluating on ten independent samples changes coverage, CV, and covering radius by at most 0.006. Convex containment follows from Proposition B.2(iv); jump counts and step lengths are measured.

Part	m	Inside	Ext. seg.	Ext. length	Depth	Ext. runs	Jumps	D_m	Cover. radius	Max. step
Cylinder	4	1.000	0	0	0	0	0	0.120	0.047	0.069
	5	1.000	0	0	0	0	0	0.065	0.023	0.039
	6	1.000	0	0	0	0	0	0.035	0.012	0.023
Sphere	4	1.000	0	0	0	0	0	0.120	0.043	0.073
	5	1.000	0	0	0	0	0	0.066	0.022	0.043
	6	1.000	0	0	0	0	0	0.036	0.012	0.025
Torus	4	1.000	< 0.001	0.001	0.029	1	0	0.179	0.039	0.205
	5	1.000	0.003	0.004	0.032	53	1	0.105	0.024	0.198
	6	0.999	0.002	0.008	0.032	241	255	0.092	0.014	0.195
L-bracket	4	0.973	0.049	0.182	0.264	53	97	0.858	0.062	0.396
	5	0.985	0.026	0.169	0.280	224	477	0.853	0.033	0.405
	6	0.993	0.012	0.157	0.289	836	2201	0.850	0.017	0.412
Complex part	4	0.988	0.038	0.058	0.045	95	0	0.197	0.041	0.166
	5	0.998	0.016	0.053	0.089	323	174	0.167	0.022	0.182
	6	0.999	0.007	0.033	0.083	1103	761	0.165	0.012	0.181

Table 6: All fifteen uniform runs, at 8^m waypoints (4096, 32 768, and 262 144). "Inside" is the fraction of waypoints inside Ω; "Ext. seg." and "Ext. length" are the fractions of exterior segments and exterior path length; "Depth" is the deepest sampled excursion. Distances (D_m, depth, covering radius, and maximum step) are in normalised units (Appendix C.1); exterior runs and jumps are counts.

Stops against the layered baseline. Table 2 gives the counts. The L-bracket's cross-sections are rectangles aligned with the raster grid, so its rows almost always continue within a layer.

C.4 Non-convex parts and refinement

These results reuse the whole-part runs and support the main-text claim that refinement reduces but does not remove the non-convex failures.

Outside centroids and crossing chords. The two non-convex mechanisms of Section 3.3 can be told apart by asking whether an exterior segment has both endpoints inside. On the L-bracket, none does at any order: every excursion starts or ends at a centroid in the region between the arms, where cells clipped into two arms place their centroids. On the complex part, a growing share have both endpoints inside (from 65 of 156 exterior segments at order four to

1218 of 1765 at order six): consecutive centroids lie on opposite sides of a hole and the chord between them crosses it. The torus lies between the two.

C.5 Decomposed source

These runs test Proposition B.3 and Corollary B.4 on the L-bracket's two-box control and on an automatic partition of the polygonal torus. No path segments are added between pieces.

Setup: partitions and solves. TetGen, through the `tetgen` 0.8.4 Python package, meshes each object once (switches `pq1.4Q`). The L-bracket is split exactly into its two boxes. For the torus, CoACD 1.0.14 [52] supplies convex hulls at threshold 0.05 and seed zero. Each tetrahedron is assigned to the hull it lies deepest inside, connected components become pieces, and components below 1% of total volume merge into a neighbour: this gives eight torus pieces. The resulting pieces need not be convex. Every tetrahedron has exactly one owner, and piece volumes conserve the total to rounding error.

All targets are uniform. Each piece receives order four, 4096 Hilbert sites, its conditional density, and time share w_j equal to its volume fraction. The unchanged pipeline and acceptance checks of Subsection C.1 are used, and all ten solves pass.

Setup: hull-gap bounds. The two boxes have $\delta_j = 0$. For each automatic piece, a branch-and-bound computation covers its convex hull by cubes, bounding the distance to the piece on each cube and refining until the upper and lower bounds U_j and L_j are within 2×10^{-3}. The computation uses floating-point geometry and winding numbers for inside tests, without an interval-arithmetic proof, so the bounds are numerical rather than exact. Their positive lower bounds show that the automatic pieces are not convex.

Results against the assigned pieces. Each segment is sampled at a spacing of at most 10^{-3} and counted exterior if any sample lies outside its piece; depth is the largest sampled distance. Table 7 reports the two cases; every sampled excursion is below its own U_j.

Object	Pieces	Waypoints	$\max_j U_j$	In piece	Exterior segments	Sampled depth
L-bracket	2	8192	0	100.00%	0.00%	0.0000
Torus	8	32 768	0.0563	98.40%	5.21%	0.0232

Table 7: Independent piece paths, uniform density, 4096 waypoints per piece. "In piece" is the percentage of waypoints inside their assigned piece; exterior segments are measured against that same piece. Depth is the largest sampled excursion; U_j is the numerical hull-gap upper bound. Distances are in normalised units (Appendix C.1).

Results against the whole object. A same-budget comparison re-evaluates two saved torus runs with 32 768 waypoints each, the whole-part order-five run and the eight-piece order-four run, against the polygonal torus. Coverage, CV, and covering radius use the same 10 000 interior samples for both; each piece waypoint is weighted by its piece's time share, $w_j/4096$, since ignoring these weights would give a misleading CV of 0.84 rather than 0.40.

Table 3 reports measurements against both the assigned pieces and the whole torus; these containment tests use different domains.

Figure run at the whole-part budget. The torus piece paths in Figure 5 come from a separate run with the same partition, pipeline, and acceptance checks at order three: eight pieces of 512 waypoints, 4096 in total, the budget of the order-four whole-part path beside them. All eight solves pass. Against the assigned pieces, 334 of the 4088 segments (8.2%) leave their own piece, with a deepest sampled excursion of 0.019, below every U_j. Against the analytic torus,

with 60 samples per segment, no segment leaves the part, whereas the order-four whole-part path has one exterior segment.

Limitations. Zero detected exterior segments is a sampled result, not a containment certificate, and the hull-gap bounds are numerical. The effect of assigned-piece crossings on local density and overlap is unmeasured, and the comparison concerns these two torus runs only.

Teaser provenance. Figure 1 uses the Stanford bunny redistributed with libigl's tutorial data (6966 triangles), without repair or simplification. TetGen (`tetgen` 0.8.4, `pq1.4Q`) meshes it once; CoACD 1.0.14 at threshold 0.1 and seed zero, followed by the tetrahedron assignment and 1% component-merge rule above, gives nine pieces. Each uniform piece receives 4096 sites, its conditional density, and a time share equal to its volume fraction, so each waypoint has global mass $w_j/4096$. All nine piece solves and the additional whole-part order-five solve pass the acceptance checks. The upper teaser row uses the whole-part path and the lower row the piece paths. Records of transforms, parameters, versions, and checks are in `results/volumetric/decomposition/`.

D Kinematics-first proof, methods, and supporting results

This appendix supplies the proof and the experimental methods behind the second application in Section 5. Subsection D.1 proves Proposition 5.1; Subsection D.2 gives the setup, measurements, and clearance results of the three-link experiment of Section 5.2.

D.1 Proof of Proposition 5.1

Proof. (i) On each of the finitely many stretches where q is continuous, the safe one-sided limits extend the lifted pair to a continuous path in F, so the stretch lies in one sheet. The projected stretch images are compact, and their union K carries all the mass; full support gives $K = \Omega$. The visited sheets therefore cover Ω, so at least c labels occur, and merging consecutive equal labels leaves at least $c - 1$ changes, as in Proposition A.5. (ii) Visiting the c patches once each gives at most $c - 1$ changes, and (i) gives at least $c - 1$. □

D.2 Three-link experimental setup

Robot, obstacles, and sheets. A planar arm with three unrestricted revolute joints and link lengths $(1, 0.5, 0.5)$ deposits on the sector $\Omega = \{(r\cos\varphi, r\sin\varphi) : 1.6 \le r \le 1.85,\ |\varphi| \le 30^\circ\}$ among three disks: two of radius 0.2 centred at $(0.9, \pm\sqrt{0.19})$ and one of radius 0.1 centred at $(1.35, 0.10)$ (Figure 7(a)). Clearance is measured for all three links against every disk. A pose counts as available, and an inverse-kinematics solution as valid, when its clearance is at least 10^{-3}. Sampling the safe fibre at four radii, $r \in \{1.6, 1.75, 1.8, 1.85\}$, on T^3 gives two endpoint-anchored families, labelled Down and Up. They are separate in the sampled fibre graph at every reported radius, and their projections overlap only for $\varphi \in [-4.4^\circ, 12.9^\circ]$. Down covers the sector up to 12.9° and Up from -4.4°, so neither covers Ω alone and $c = 2$ in the sampled model.

Chart, waypoints, and densities. Both pipelines use the same intended sector chart, $r^2 = 1.6^2 + s(1.85^2 - 1.6^2)$, $\varphi = -30^\circ + 60^\circ v$, $(s, v) \in [0, 1]^2$. It is area-preserving up to a constant, so uniform source density in (s, v) gives uniform area density. Both use the same 4^m order-m Hilbert waypoints at orders $m = 2, 3, 4$, and two target densities: uniform, and a gradient density that relative to area rises linearly in v from 0.5 to 1.5. The gradient is imposed analytically by the inverse cumulative map $v \mapsto -\frac{1}{2} + \sqrt{\frac{1}{4} + 2v}$. Waypoints are joined by straight chart segments, subdivided into collocation points no more than 0.004 apart in the workspace.

Geometry-first baseline. The baseline follows the Hilbert path over the whole chart, solves constrained inverse kinematics at each collocation point, warm-started from the previous solution, and changes posture only when the current family becomes unavailable. Solving afresh at every point is a property of this baseline, not of the geometry-first order: a geometry-first planner could also cache inverse maps.

Kinematics-first preparation. Preparation builds two patches: Down over $\varphi \in [-30°, 12.5°]$ and Up over $[-4°, 30°]$, both over the whole radial interval. For each patch, it solves inverse kinematics in the selected family on a 17×129 grid of the patch chart, with clearance at least 0.004 at the vertices. It stores the result as a piecewise-affine joint-pose map on the triangulated grid, and certifies every one of its 4096 joint-space triangles collision-free. Then forward kinematics of the stored map is compared with the intended chart on a denser 33×257 grid. The largest deviation is 0.0018 for Down and 0.00096 for Up, against an acceptance threshold of 0.002. This is a sampled validation tolerance, not a certified bound on tracking error over the patch.

Kinematics-first requests. Each request splits the source square at the target cut 4.25°, the middle of the sampled overlap. Whole Hilbert rows are assigned to each patch so that their share is the closest available to that patch's share of target mass. Rounding changes the share: for uniform density at order four, 0.571 becomes 0.5625. The v-coordinates of each group are rescaled so that the row boundary maps to the target cut, the density map is applied, and the result is expressed in that patch's chart. The stored map then gives joint poses without any inverse-kinematics solve. Collocation points include every crossing of a triangle edge, so the joint path, linear between them, equals the stored map along each chart segment. The tool path is not prescribed. It is FK of this joint motion, and only approximates the intended chart: across all requests, the largest sampled deviation at collocation points is 9.6×10^{-4}.

The construction realises the theoretical lifts σ_i only approximately. Write ξ_i for the source path of patch i and $\Psi_{k,i}$ for the composition of rescaling, density map, patch chart, and stored joint-pose map. The implementation is then

$$q = \Psi_{k,i} \circ \xi_i, \qquad x = \mathrm{FK} \circ q,$$

an instance of (10), with $\mathrm{FK} \circ \Psi_{k,i}$ approximating the intended chart rather than equal to it. Parameterising a prepared lift through workspace coordinates does not in itself determine the construction order. What makes this construction kinematics-first is that joint motion comes from the source through a prepared, certified map, and the tool path is derived from it. The prescribed tool path is not lifted point by point. The density condition of Section 5 is targeted through the analytic chart and density map. A small position error does not establish it, because a small deviation in x does not bound the deviation of the Jacobian J_S.

The implementation does not generate a separate Hilbert traversal for each patch. It restricts the one order-m Hilbert sequence to the two row groups and concatenates them. The two groups are joined through a common handoff point in the overlap, where one posture reset occurs (Figure 7(b,d)). A restricted sequence keeps the original order of its points, but where the full curve leaves the group and re-enters, consecutive retained points can be far apart, so these restrictions can introduce longer connections. The connectors to and from the handoff point add tool travel outside the waypoint budget.

Certification. For both pipelines, every continuous stretch of the joint path is certified collision-free over adaptively subdivided pieces, using a Lipschitz lower bound on clearance. The certificate establishes positive clearance; it does not establish the 10^{-3} margin. A *reset* is a discrete posture change at a fixed tool point. It is counted, but the physical transition between postures is neither planned nor certified.

Measurements. The *switch count* is the number of observed resets. The *coverage error* is the largest distance from a 33×129 target grid to the sampled depositing path. The *density error* is the total-variation distance between target shares and the shares of the original waypoints, mapped through FK, on 4×8 area bins. It measures where waypoints land, not continuous deposition over time: connectors, tool speed, and material flow are not in the score. At order two, 16 waypoints cannot fill 32 bins, so the uniform density error cannot fall below 0.5. Timing uses wall-clock medians on one machine with an unoptimised Python implementation, after one untimed warm-up request. A shared fibre analysis (8.2 s) serves both pipelines and is included in the cumulative times reported in Section 5.2. *Preparation* is the method-specific one-time cost, and *planning* is the time to plan and certify one request. Requests change the Hilbert order or the target density while the robot, obstacles, and target stay fixed. Kinematics-first retains its prepared maps across requests. The geometry-first baseline retains only the shared fibre analysis: as implemented, its inverse-kinematics solves and certificates belong to each request's tool points.

Clearance. Certified clearance bounds are positive but conservative worst-piece values, far below the sampled minima. Sampled minimum clearance is 0.0038 to 0.0040 for kinematics-first. For geometry-first, it is 0.00077 to 0.00086, below the 10^{-3} imposed at each collocation point, so the geometry-first paths do not keep that margin between points.

References

[1] Guoxin Fang, Tianyu Zhang, Yuming Huang, Zhuo Zhang, Kunal Masania, and Charlie C. L. Wang. Exceptional mechanical performance by spatial printing with continuous fiber: Curved slicing, toolpath generation and physical verification. *Additive Manufacturing*, 82:104048, 2024.

[2] Donghong Ding, Zengxi Pan, Dominic Cuiuri, and Huijun Li. A tool-path generation strategy for wire and arc additive manufacturing. *The International Journal of Advanced Manufacturing Technology*, 73(1–4):173–183, 2014. doi: 10.1007/s00170-014-5808-5.

[3] Yi Wei Daniel Tay, Guan Heng Andrew Ting, Ye Qian, Biranchi Panda, Lewei He, and Ming Jen Tan. Time gap effect on bond strength of 3D-printed concrete. *Virtual and Physical Prototyping*, 14 (1):104–113, 2019. doi: 10.1080/17452759.2018.1500420.

[4] Ben Ezair, Saul Fuhrmann, and Gershon Elber. Volumetric covering print-paths for additive manufacturing of 3D models. *Computer-Aided Design*, 100:1–13, 2018.

[5] Chengkai Dai, Charlie C. L. Wang, Chenming Wu, Sylvain Lefebvre, Guoxin Fang, and Yong-Jin Liu. Support-free volume printing by multi-axis motion. *ACM Transactions on Graphics*, 37(4): 134:1–134:14, 2018. doi: 10.1145/3197517.3201342.

[6] Xavier Chermain, Giovanni Cocco, Cédric Zanni, Eric Garner, Pierre-Alexandre Hugron, and Sylvain Lefebvre. Atomizer: Beyond non-planar slicing for fused filament fabrication. *Computer Graphics Forum*, 44:e70189, 2025.

[7] Neelotpal Dutta, Tianyu Zhang, Tao Liu, Yongxue Chen, and Charlie C. L. Wang. Implicit neural field-based process planning for multi-axis manufacturing: Direct control over collision avoidance and toolpath geometry, 2025. arXiv:2511.17578.

[8] Tong Yang, Jaime Valls Miro, Yue Wang, and Rong Xiong. Non-revisiting coverage task with minimal discontinuities for non-redundant manipulators. In *Proceedings of Robotics: Science and Systems*, Corvallis, Oregon, USA, July 2020. doi: 10.15607/RSS.2020.XVI.005.

[9] Tianyu Zhang, Xiangjia Chen, Guoxin Fang, Yingjun Tian, and Charlie C. L. Wang. Singularity-aware motion planning for multi-axis additive manufacturing, 2021. arXiv:2103.00273.

[10] Tianyu Zhang, Guoxin Fang, Yuming Huang, Neelotpal Dutta, Sylvain Lefebvre, Zekai Murat Kilic, and Charlie C. L. Wang. S^3-Slicer: A general slicing framework for multi-axis 3D printing. *ACM Transactions on Graphics*, 41(6):277:1–277:15, 2022. doi: 10.1145/3550454.3555516.

[11] Tao Liu, Tianyu Zhang, Yongxue Chen, Yuming Huang, and Charlie C. L. Wang. Neural slicer for multi-axis 3D printing. *ACM Transactions on Graphics*, 43(4):85:1–85:15, 2024. doi: 10.1145/3658212.

[12] Prusa Research. PrusaSlicer: open-source slicer, Spiral Vase mode. Software, 2026. URL `https://github.com/prusa3d/PrusaSlicer`. Accessed 7 October 2026.

[13] A. Dolenc and I. Mäkelä. Slicing procedures for layered manufacturing techniques. *Computer-Aided Design*, 26(2):119–126, 1994. doi: 10.1016/0010-4485(94)90032-9.

[14] Pulak Mohan Pandey, N. Venkata Reddy, and Sanjay G. Dhande. Slicing procedures in layered manufacturing: a review. *Rapid Prototyping Journal*, 9(5):274–288, 2003. doi: 10.1108/13552540310502185.

[15] Marc Alexa, Kristian Hildebrand, and Sylvain Lefebvre. Optimal discrete slicing. *ACM Transactions on Graphics*, 36(1):12:1–12:16, 2017. doi: 10.1145/2999536.

[16] Haisen Zhao, Fanglin Gu, Qi-Xing Huang, Jorge Garcia, Yong Chen, Changhe Tu, Bedrich Benes, Hao Zhang, Daniel Cohen-Or, and Baoquan Chen. Connected Fermat spirals for layered fabrication. *ACM Transactions on Graphics*, 35(4):100:1–100:10, 2016. doi: 10.1145/2897824.2925958.

[17] Antoine Bedel, Yoann Coudert-Osmont, Jonàs Martínez, Rahnuma Islam Nishat, Sue Whitesides, and Sylvain Lefebvre. Closed space-filling curves with controlled orientation for 3D printing. *Computer Graphics Forum*, 41(2):473–492, 2022. doi: 10.1111/cgf.14488.

[18] Prashant Gupta, Bala Krishnamoorthy, and Gregory Dreifus. Continuous toolpath planning in additive manufacturing, 2019. arXiv:1908.07452.

[19] Jimmy Etienne, Nicolas Ray, Daniele Panozzo, Samuel Hornus, Charlie C. L. Wang, Jonàs Martínez, Sara McMains, Marc Alexa, Brian Wyvill, and Sylvain Lefebvre. CurviSlicer: Slightly curved slicing for 3-axis printers. *ACM Transactions on Graphics*, 38(4):81:1–81:11, 2019. doi: 10.1145/3306346.3323022.

[20] Maxime Chalvin, Sébastien Campocasso, Vincent Hugel, and Thomas Baizeau. Layer-by-layer generation of optimized joint trajectory for multi-axis robotized additive manufacturing of parts of revolution. *Robotics and Computer-Integrated Manufacturing*, 65:101960, 2020. doi: 10.1016/j.rcim.2020.101960.

[21] Markus Schmitz, Jan Wiartalla, Markus Gelfgren, Samuel Mann, Burkhard Corves, and Mathias Hüsing. A robot-centered path-planning algorithm for multidirectional additive manufacturing for WAAM processes and pure object manipulation. *Applied Sciences*, 11(13):5759, 2021. doi: 10.3390/app11135759.

[22] Fanchao Zhong, Yonglai Xu, Haisen Zhao, and Lin Lu. As-continuous-as-possible extrusion-based fabrication of surface models. *ACM Transactions on Graphics*, 42(3):26, 2023. doi: 10.1145/3575859.

[23] Donghong Ding, Zengxi Pan, Dominic Cuiuri, and Huijun Li. A practical path planning methodology for wire and arc additive manufacturing of thin-walled structures. *Robotics and Computer-Integrated Manufacturing*, 34:8–19, 2015. doi: 10.1016/j.rcim.2015.01.003.

[24] Adama Diourté, Florian Bugarin, Cyril Bordreuil, and Stéphane Segonds. Continuous three-dimensional path planning (CTPP) for complex thin parts with wire arc additive manufacturing. *Additive Manufacturing*, 37:101622, 2021.

[25] David Hilbert. Über die stetige Abbildung einer Linie auf ein Flächenstück. *Mathematische Annalen*, 38:459–460, 1891.

[26] Hans Sagan. *Space-Filling Curves*. Universitext. Springer-Verlag, New York, 1994. doi: 10.1007/978-1-4612-0871-6.

[27] Michael Bader. *Space-Filling Curves: An Introduction with Applications in Scientific Computing*, volume 9 of *Texts in Computational Science and Engineering*. Springer, Berlin, Heidelberg, 2013. doi: 10.1007/978-3-642-31046-1.

[28] Herman Haverkort. How many three-dimensional Hilbert curves are there? *Journal of Computational Geometry*, 8(1):206–281, 2017. arXiv:1610.00155.

[29] Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on Pure and Applied Mathematics*, 44(4):375–417, 1991.

[30] Gabriel Peyré and Marco Cuturi. *Computational optimal transport: With applications to data science*. now Publishers, 2019. doi: 10.1561/2200000073.

[31] M. Bertoldi, M. A. Yardimci, C. M. Pistor, and S. I. Guceri. Domain decomposition and space filling curves in toolpath planning and generation. In *Proceedings of the Solid Freeform Fabrication Symposium*, pages 267–274, 1998.

[32] Alexios Papacharalampopoulos, Harry Bikas, and Panagiotis Stavropoulos. Path planning for the infill of 3D printed parts utilizing Hilbert curves. *Procedia Manufacturing*, 21:757–764, 2018.

[33] Luis A. Caffarelli. The regularity of mappings with a convex potential. *Journal of the American Mathematical Society*, 5(1):99–104, 1992.

[34] Yash Jhaveri and Ovidiu Savin. On the regularity of optimal transports between degenerate densities. *Archive for Rational Mechanics and Analysis*, 245:819–861, 2022. doi: 10.1007/s00205-022-01796-y. URL `https://arxiv.org/abs/2105.04312v2`. Published manuscript: arXiv:2105.04312v2.

[35] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998. doi: 10.1007/PL00009187.

[36] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5): 1583–1592, 2011. doi: 10.1111/j.1467-8659.2011.02032.x.

[37] Bruno Lévy. A numerical algorithm for L_2 semi-discrete optimal transport in 3D. *ESAIM: Mathematical Modelling and Numerical Analysis*, 49(6):1693–1715, 2015. doi: 10.1051/m2an/2015055.

[38] Jun Kitagawa, Quentin Mérigot, and Boris Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society*, 21(9):2603–2651, 2019. doi: 10.4171/JEMS/889.

[39] Hang Si. TetGen, a Delaunay-based quality tetrahedral mesh generator. *ACM Transactions on Mathematical Software*, 41(2):11:1–11:36, 2015. doi: 10.1145/2629697.

[40] Yijiang Huang, Caelan Reed Garrett, and Caitlin T. Mueller. Automated sequence and motion planning for robotic spatial extrusion of 3D trusses, 2018. arXiv:1810.00998.

[41] Philippe Wenger and Patrick Chedmail. Ability of a robot to travel through its free work space in an environment with obstacles. *The International Journal of Robotics Research*, 10(3):214–227, 1991.

[42] Paul Borrel and Alain Liégeois. A study of multiple manipulator inverse kinematic solutions with applications to trajectory planning and workspace determination. In *Proceedings of the 1986 IEEE International Conference on Robotics and Automation*, volume 3, pages 1180–1185. IEEE, 1986.

[43] Steven M. LaValle. *Planning Algorithms*. Cambridge University Press, 2006.

[44] Joel W. Burdick. On the inverse kinematics of redundant manipulators: characterization of the self-motion manifolds. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 264–270, 1989.

[45] Daniel H. Gottlieb. Robots and fibre bundles. *Bulletin de la Société Mathématique de Belgique*, 38: 219–223, 1988.

[46] Petar Pavešić. A topologist's view of kinematic maps and manipulation complexity, 2017. arXiv:1707.03899.

[47] Kazem Kazerounian and Zhaoyu Wang. Global versus local optimization in redundancy resolution of robotic manipulators. *The International Journal of Robotics Research*, 7(5):3–12, 1988. doi: 10.1177/027836498800700501.

[48] Thierry Champion, Luigi De Pascale, and Petri Juutinen. The ∞-Wasserstein distance: local solutions and existence of optimal transport maps. *SIAM Journal on Mathematical Analysis*, 40(1): 1–20, 2008. doi: 10.1137/07069938X.

[49] Guido De Philippis and Alessio Figalli. Partial regularity for optimal transport maps. *Publications mathématiques de l'IHÉS*, 121:81–112, 2015.

[50] Quentin Mérigot, Alex Delalande, and Frédéric Chazal. Quantitative stability of optimal transport maps and linearization of the 2-Wasserstein space. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 108 of *Proceedings of Machine Learning Research*, pages 3186–3196, 2020.

[51] John Skilling. Programming the Hilbert curve. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, volume 707 of *AIP Conference Proceedings*, pages 381–387, 2004. doi: 10.1063/1.1751381.

[52] Xinyue Wei, Minghua Liu, Zhan Ling, and Hao Su. Approximate convex decomposition for 3D meshes with collision-aware concavity and tree search. *ACM Transactions on Graphics*, 41(4): 42:1–42:18, 2022. doi: 10.1145/3528223.3530103. URL `https://github.com/SarahWeiii/CoACD`.